

DEVELOPMENT OF FINITE ELEMENT SOLVER FOR LIGHT SCATTERING SIMULATION

LEE SHIEN WEI

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ABSTRACT

This project addresses a critical implementation gap in computational electromagnetics by developing an open-source finite element solver for light scattering simulation using high-order Nédélec edge elements within the FEniCSx framework. The first objective focused on solver development and stabilisation, to implement a high-accuracy finite element solver utilising unstructured meshes and vector edge elements to ensure physical correctness and geometric flexibility. The methodology employs third-order $H(\text{curl})$ -conforming Nédélec elements on unstructured triangular meshes to discretise the time-harmonic vector Helmholtz equation. The second objective was to verify the solver accuracy against the analytical Mie series benchmark for a two-dimensional perfect electric conductor cylinder. Quantitative validation revealed that the solver operates within the stable asymptotic regime where $kh = 0.374$, avoiding numerical pollution effects. The mesh convergence study provided definitive proof of implementation correctness, demonstrating that the numerical error decreases monotonically with mesh refinement and achieves an empirical convergence rate of 2.88, in excellent agreement with the theoretical prediction of 3.0 for cubic Nédélec elements. The third objective was to validate robust domain truncation techniques to simulate an infinite free-space environment within a limited computational grid, specifically progressing from basic Absorbing Boundary Conditions (ABC) to advanced Anisotropic Perfectly Matched Layers (APML) to completely eliminate artificial boundary reflections. Upgrading the domain truncation boundary to an Anisotropic Perfectly Matched Layer successfully eliminated boundary-induced reflection noise, reducing the final global relative error to 0.019%. When deployed to model a penetrable 25 nm radius gold core-shell nanoparticle, the volume-source formulation accurately tracked the macroscopic optical transduction, yielding a highly linear bulk sensitivity of 321.64 nm/RIU with a coefficient of determination of $R^2 = 0.99523$. Furthermore, the parametric surface decay analysis extracted a characteristic evanescent confinement length of exactly 5.1 nm, providing rigorous data-driven validation of the near-field plasmonic physics. This research directly advances United Nations Sustainable Development Goals by democratising professional-grade computational tools for researchers and students without advanced programming skills, and lowering financial barriers to support global scientific innovation and equitable engineering education.

ABSTRAK

Projek ini menangani jurang pelaksanaan kritikal dalam elektromagnetik pengiraan dengan membangunkan penyelesaian unsur terhingga sumber terbuka untuk simulasi penyerakan cahaya menggunakan elemen tepi Nédélec bertertib tinggi dalam rangka kerja FEniCSx. Objektif pertama berfokus pada pembangunan dan pemantapan penyelesaian untuk melaksanakan penyelesaian unsur terhingga ketepatan tinggi memanfaatkan jaringan tidak berstruktur dan elemen tepi vektor bagi memastikan ketepatan fizikal dan kefleksibelan geometri. Metodologi ini menggunakan elemen Nédélec pematuhan $H(\text{curl})$ bertertib ketiga pada jaringan segi tiga tidak berstruktur untuk mendiskritkan persamaan Helmholtz vektor harmonik masa. Objektif kedua adalah untuk mengesahkan ketepatan penyelesaian terhadap penanda aras siri Mie analitikal untuk silinder konduktor elektrik sempurna dua dimensi. Pengesahan kuantitatif mendedahkan bahawa penyelesaian beroperasi dalam rejim asimptotik stabil di mana $kh = 0.374$, bagi mengelakkan kesan pencemaran berangka. Kajian penumpuan menunjukkan ralat berkurangan secara monotonik merentasi lima peringkat penghalusan jaringan dan mencapai kadar penumpuan empirik 2.88, yang sangat sepadan dengan ramalan teori 3.0 untuk elemen Nédélec kubik. Objektif ketiga adalah untuk mengesahkan teknik pemotongan domain yang teguh, khususnya syarat sempadan penyerapan untuk mensimulasikan persekitaran ruang bebas tidak terikat dan melanjutkan konsep ini kepada aplikasi biosensor resonans plasmon permukaan setempat. Menaik taraf pemotongan domain kepada Lapisan Padanan Sempurna Anisotropik berjaya menghapuskan ralat pantulan sempadan, sekali gus mengurangkan ralat relatif global akhir kepada 0.019%. Apabila digunakan pada nanozarah teras cangkerang emas berjajari 25 nm, penyelesaian berjaya memetakan transduksi optik makroskopik dengan mencatatkan kepekaan pukal linear sebanyak 321.64 nm/RIU dengan pekali penentuan $R^2 = 0.99523$. Selain itu, analisis perebutan permukaan fana berjaya mengekstrak panjang pembatasan fana yang tepat iaitu 5.1 nm, yang mengesahkan fizik plasmonik medan dekat. Penyelidikan ini memajukan Matlamat Pembangunan Mampan Pertubuhan Bangsa-Bangsa Bersatu dengan mendemokrasikan alat pengiraan gred profesional untuk penyelidik tanpa kemahiran pengaturcaraan lanjutan, serta menghapuskan halangan kewangan bagi menyokong inovasi saintifik yang mampan dan pendidikan kejuruteraan yang saksama.

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LIST OF SYMBOLS AND ABBREVIATIONS

Symbols

E	Electric field intensity
H	Magnetic field intensity
D	Electric flux density
B	Magnetic flux density
J	Conductive electric current density
ρ	Volume electric charge density
ϵ	Permittivity
μ	Permeability
σ	Conductivity
ϵ_r	Relative permittivity
μ_r	Relative permeability
k_0	Free space wavenumber
ω	Angular frequency
λ	Wavelength
h	Mesh size / average element size
kh	Dimensionless resolution parameter
ka	Size parameter (resonance regime)
p	Polynomial degree/order of basis functions
Ω	Computational domain/volume
$\partial\Omega$	Domain boundary
$\hat{n}$	Unit normal vector

Abbreviations

ABC	Absorbing Boundary Condition
APML	Anisotropic Perfectly Matched Layer
BEM	Boundary Element Method
DOF	Degree of Freedom
FDTD	Finite Difference Time Domain
FDFD	Finite Difference Frequency Domain
FEM	Finite Element Method
HPC	High-Performance Computing
LOD	Limit of Detection
LSPR	Localized Surface Plasmon Resonance
LU	Lower-Upper (Factorization/Decomposition)
MPI	Message Passing Interface
MUMPS	MUltifrontal Massively Parallel sparse direct Solver
PEC	Perfect Electric Conductor
PETSc	Portable, Extensible Toolkit for Scientific Computation
PMC	Perfect Magnetic Conductor
PML	Perfectly Matched Layer
SDG	Sustainable Development Goal
TE	Transverse Electric (mode)
UFL	Unified Form Language

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Appendix A	Turnitin Similarity Level
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Chapter 1

Introduction

1.1 Introduction to Light Scattering

Light scattering is a ubiquitous and fundamental physical phenomenon that describes the interaction of electromagnetic waves with discrete particles or inhomogeneous media. Whether it is the blue color of the sky, the brilliance of a white cloud, or the operation of radar systems, scattering occurs whenever an incident electromagnetic wave encounters an obstacle. This interaction induces oscillating multipoles, described as microscopic electric charge separations that vary with time within the object, which subsequently radiate secondary waves known as the scattered field. Understanding how the amplitude, phase and polarization of light evolve during this interaction is essential for a wide array of scientific and engineering disciplines.

The detailed study of these interactions is a cornerstone for the development of advanced technologies. In remote sensing, the analysis of scattered waves allows for the detection and characterization of objects that are otherwise inaccessible. Similarly, in biomedical optics, the scattering properties of tissues are exploited to diagnose diseases at an early stage. However, the complexity of practical scatterers, which often possess irregular shapes and inhomogeneous material properties, renders analytical solutions such as the Mie series insufficient (Wriedt, 2012). While Mie theory provides an exact mathematical solution for scattering by simple spheres and has served as the gold standard for validation in the field, it cannot be applied to the arbitrary geometries encountered in practical engineering problems. Consequently, numerical simulation has become an indispensable tool. By discretizing the governing equations of electromagnetism and converting the continuous differential equations into a system of algebraic equations solvable by computer, numerical methods allow researchers to predict scattering behavior with high fidelity (Rumpf et al., 2014).

The physics of this interaction is governed entirely by the laws of classical electromag-

netism. At the macroscopic level, the behavior of these electromagnetic fields is described by a set of coupled partial differential equations known as Maxwell's equations.

1.1.1 Maxwell's Equations

In the context of light scattering, the fundamental behavior of electromagnetic fields is described by Maxwell's equations. For a linear, isotropic medium characterized by permittivity ϵ (a measure of electric polarization), permeability μ (a measure of magnetic response) and conductivity σ , the general time-domain differential forms of these equations are:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (1.1)$$

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} + \mathbf{J} \quad (1.2)$$

$$\nabla \cdot \mathbf{D} = \rho \quad (1.3)$$

$$\nabla \cdot \mathbf{B} = 0 \quad (1.4)$$

where $\mathbf{D} = \epsilon \mathbf{E}$ is the electric flux density and $\mathbf{B} = \mu \mathbf{H}$ is the magnetic flux density.

Faraday's Law of Induction (1.1).

This equation describes the fundamental principle of electromagnetic induction. It states that a time-varying magnetic flux density ($\mathbf{B}$) induces a spatially circulating electric field intensity ($\mathbf{E}$) (Jin, 2015). Physically, this coupling implies that a changing magnetic environment is capable of generating an electromotive force, which is the mechanism governing devices like transformers and generators (Jin, 2015).

Ampère's Circuital Law (1.2).

This law establishes that the curl of the magnetic field intensity ($\mathbf{H}$) is generated by two distinct sources which are the flow of conductive electric current density ($\mathbf{J}$) and the time rate of change of the electric flux density ($\mathbf{D}$) (Jin, 2015). The latter term, $\frac{\partial \mathbf{D}}{\partial t}$, represents the "displacement current," which is critical for explaining how magnetic fields can exist in regions without physical charge flow (such as vacuum or dielectrics), thereby allowing for the propagation of electromagnetic waves (Jin, 2015).

Gauss's Law for Electric Fields (1.3).

Gauss's law for electric fields states that the divergence of the electric flux density ($\mathbf{D}$) is directly proportional to the volume electric charge density (ρ) (Jin, 2015). Physically, this signifies that

electric charges act as the sources (positive charge) or sinks (negative charge) of the electric field where the flux lines diverge outwards from positive charge and converge inwards towards negative charge (Jin, 2015).

Gauss's Law for Magnetic Fields (1.4).

This equation asserts that the divergence of the magnetic flux density ($\mathbf{B}$) is always zero (Jin, 2015). Physically, this implies the non-existence of isolated magnetic charges (magnetic monopoles). Consequently, magnetic field lines are solenoidal, meaning they always form continuous, closed loops with no distinct beginning or end (Jin, 2015).

1.1.2 Significance of Light Scattering Simulation

The simulation of light scattering has become indispensable in biomedical engineering driven by the critical need for non-invasive diagnostic tools and the ethical constraints associated with direct human or animal experimentation (Periyasamy and Pramanik, 2017; Krasnikov et al., 2019). Recognized as the gold standard for modeling light transport, Monte Carlo (MC) simulations allow researchers to rigorously study photon interactions within complex, heterogeneous biological structures which ranging from the intricate layers of the human eye to whole-body small animal models where traditional analytical solutions often fail (Periyasamy and Pramanik, 2017; Krasnikov et al., 2019). The implementation of these methods has been revolutionized by the integration of realistic mesh-based geometries and parallel processing technologies like Graphics Processing Units (GPUs), which address the massive computational demands required to trace millions of photon paths for statistical accuracy (Periyasamy and Pramanik, 2017; Krasnikov et al., 2019). Consequently, these advanced simulations provide a robust platform for optimizing novel imaging systems, such as Optical Coherence Tomography (OCT) and Raman spectroscopy and are essential for quantifying light distribution to ensure safety in therapeutic applications like optogenetics without the risks of invasive procedures (Krasnikov et al., 2019).

Parallel to these advancements in the biomedical domain, light scattering simulation stands as a foundational pillar in the rapidly evolving field of nanophotonics, where modern systems involve intricate light-matter interaction regimes that extend well beyond the diffraction limit (Gallinet et al., 2015). As research progresses from simple plasmonic nanostructures to increasingly sophisticated multiscale devices, traditional analytical methods become insufficient, rendering numerical modeling central to the understanding, control and design of these complex systems (Gallinet et al., 2015). Such simulations are particularly indispensable for engineering exotic metamaterials including sub-wavelength focusing lenses and for optimizing photonic bandgap structures, where the precise manipulation of light propagation, localization and scattering is critical for performance (Teixeira et al., 2023). Consequently, the continuous

demand for highly efficient numerical tools is driven by the necessity to accurately model these elaborate physical phenomena, thereby bridging the gap between theoretical nanophotonic concepts and the engineering of functional optical devices for applications ranging from sensing and light harvesting to optical signal processing (Gallinet et al., 2015; Teixeira et al., 2023).

Nanoplasmonic biosensors utilizing localized surface plasmon resonance have emerged as a premier archetype for modern label-free clinical diagnostics (Mayer and Hafner, 2011). These optical systems exploit the collective oscillation of free conduction electrons in noble metal nanostructures when excited by the oscillating electric field of an incident electromagnetic wave (Mayer and Hafner, 2011). This light-matter interaction generates sharp spectral absorption and scattering peaks alongside intense electromagnetic near-field enhancements that are highly concentrated at the metallic surface and decay rapidly with distance (Mayer and Hafner, 2011; Saini et al., 2023). Among various configurations, core-shell nanostructures comprising a solid metallic core and an outer biological receptor layer represent a vital model for investigating near-field optical physics and evaluating sensor optimization strategies (Mayer and Hafner, 2011). Gold nanoparticles synthesized through established chemical routes like the Turkevich method serve as an excellent material core for these assemblies due to their high chemical stability and resistance to oxidation (Saini et al., 2023).

In practical biological applications, the thin outer shell of the nanoparticle is functionalized with specific bioreceptors or capture agents designed to isolate target medical biomarkers (Mayer and Hafner, 2011). When target biological analytes bind within this capture matrix, they alter the local ambient refractive index within the highly confined evanescent sensing volume (Mayer and Hafner, 2011; Clementi et al., 2019). According to classical scattering theory, the total optical extinction cross-section is maximized when the real part of the complex dielectric function of the metal satisfies a specific resonance condition balancing the dielectric constant of the surrounding medium (Mayer and Hafner, 2011; Clementi et al., 2019). Localized refractive index shifts modify these boundary conditions, forcing the resonance state to shift toward longer wavelengths, which produces a measurable macroscopic redshift in the extinction peak (Mayer and Hafner, 2011). Because the field enhancement decays exponentially, the thickness of the bioreceptor capture layer must be minimized and strictly kept below 20 nm to prevent premature signal saturation and ensure that the spectral shift remains detectable (Mayer and Hafner, 2011).

Accurately predicting these nanoplasmonic interactions requires rigorous computational frameworks capable of resolving complex nanoscale optical physics at deforming material boundaries (Saini et al., 2023). Researchers widely utilize volumetric numerical approaches like the finite element method in the frequency domain to solve the governing time-harmonic Maxwell equations and map the scattered electric field distributions (Saini et al., 2023). In the

long wavelength limit where the incident wave is significantly larger than the nanoparticle size, the mathematical model can be safely approximated using electrostatic or quasistatic theory to analyze complex geometric interactions at a lower computational cost (Clementi et al., 2019). For deep sub-wavelength particles smaller than 20 nm, absorption processes heavily dominate the extinction profile while scattering contributions become negligible (Clementi et al., 2019). By executing automated parametric sweeps that systematically alter structural dimensions and evaluating metrics like bulk sensitivity slopes and peak linewidth figures of merit, researchers can optimize device topographies to maximize the optical signal (Mayer and Hafner, 2011; Saini et al., 2023). These high-fidelity simulations can even capture sub-nanometer shifts such as a precise 0.5 nm resonance shift caused by individual proteins binding near the metallic boundary, directly supporting the development of ultra-sensitive label-free diagnostic devices (Clementi et al., 2019).

1.2 Problem Statement

The precise simulation of light scattering is a cornerstone in the development of advanced nanophotonic and biomedical devices, where functionality often hinges on the interaction between electromagnetic waves and complex, sub wavelength structures. Historically, analytical techniques like the Mie series have served as the primary tool for such analysis (Wriedt, 2012). However, as detailed by Martin (1993), the applicability of these formulae is rigorously constrained to perfect spheres or infinite cylinders. This introduces significant errors not only when applied to the anisotropic, swelling microstructures found in hydrogel biosensors but also when modeling irregular biological organelles or non-canonical nanostructures. To address these geometric limitations, numerical partial differential equation solvers are utilized, though each approach presents distinct trade-offs. The Finite Difference Time Domain (FDTD) method is often cited for its versatility; Drezek et al. (2000) demonstrate its utility in modeling light scattering from heterogeneous biological cells, where it offers greater geometric flexibility than Mie theory. Furthermore, Teixeira et al. (2023) emphasize FDTD’s robustness in handling broad spatial scales and multiphysics scenarios. However, both FDTD and the Finite Difference Frequency Domain (FDFD) method rely on orthogonal Cartesian grids. As Shin (2013) highlights, this discretization inherently suffers from “staircasing” errors when modeling curved interfaces, potentially leading to inaccurate near field predictions for plasmonic nanostructures or deforming sensor surfaces. Similarly, while Boundary Element Methods (BEM) provide rigorous solutions for radiation problems (Cools et al., 2011), they can become computationally expensive when applied to penetrable objects with highly inhomogeneous material properties, such as biological tissues or gradient index lenses (Buffa and Hiptmair, 2003). In this context, the Finite Element Method (FEM) offers a distinct advantage for modeling these complex, curved boundaries through the use of unstructured tetrahedral meshes, which naturally conform to both the

deforming geometry of biosensors and the intricate morphology of cellular structures.

The choice between time domain and frequency domain formulations depends heavily on the specific optical setup and the nature of the excitation source. For applications requiring broadband spectral analysis such as analyzing the scattering cross section of a nanoparticle over the entire visible spectrum time domain methods like finite element time domain method are advantageous as they resolve the frequency response in a single simulation run (Teixeira et al., 2023). However, for specific classes of devices interrogated by monochromatic light, the frequency domain numerical approach becomes computationally more efficient. This is particularly relevant for the stimuli responsive biosensors discussed in this study, as well as for designing specialized photonic crystals and metamaterials operating at fixed laser wavelengths. Using a time domain numerical solver for such steady state, monochromatic problems require simulating wave propagation over thousands of time steps until convergence is achieved, a process Drezek et al. (2000) note can be resource intensive. In contrast, a frequency domain numerical solver provides a direct, one-shot solution to the time harmonic Maxwell's equations, matching the continuous wave nature of the excitation. To ensure accuracy in this regime, however, the solver must be carefully designed to avoid artificial errors. Standard nodal based finite element methods often can violate the continuity of tangential electric fields across dielectric boundaries, leading to spurious solutions. Schwarzbach (2009) demonstrates that this flaw frequently leads to spurious solutions that look like real signals but are numerical noise. To prevent this, it is essential to use high order Nédélec (edge) elements, which enforce the correct physical continuity of the electric field, thereby distinguishing true electromagnetic effects (Koba and Szczepanski, 2012) from simulation errors.

Beyond the challenges of complex geometry and software implementation, simulating practical nanophotonic devices like Localized Surface Plasmon Resonance (LSPR) biosensors introduces significant physical and computational hurdles. The first major barrier is the severe optical dispersion of real plasmonic metals, such as gold, whose material properties change drastically and continuously across different light wavelengths. Relying on static approximations for these materials completely fails to capture the true resonant behaviour required for clinical diagnostics. The second major barrier involves mathematical instabilities within High-Performance Computing (HPC) environments. When researchers attempt to calculate how light scatters by evaluating surface flux integrals across partitioned mesh boundaries, it frequently triggers severe system crashes and communication bottlenecks under parallel processing. Consequently, this project aims to resolve these advanced nanoplasmonic challenges by integrating continuous cubic-spline interpolations to accurately model the dispersive metal properties. Furthermore, it employs an inhomogeneous volume-source formulation that leverages the Rayleigh scattering approximation. This strategic physical approximation effectively bypasses the unstable surface calculations, ensuring robust computational stability without sacrificing the tracking

precision needed for modern biosensor design.

1.3 Research Objectives

The overarching goal of this research is to develop and validate a robust finite element solver for light scattering simulations. The specific objectives are:

- **Mathematical Formulation and Approximation:** To formulate the variational form of the time harmonic equation.
- **Solver Development and Stabilization:** To implement a high-accuracy finite element solver utilising unstructured meshes and vector edge elements (Nédélec elements) (Nedelec, 1980) to ensure physical correctness and geometric flexibility.
- **Domain Truncation and Validation:** To validate robust domain truncation techniques to simulate an infinite free-space environment within a limited computational grid, specifically progressing from basic Absorbing Boundary Conditions (ABC) to advanced Anisotropic Perfectly Matched Layers (APML) to completely eliminate artificial boundary reflections.

1.4 Scope of Work

This project focuses on developing a finite element solver for light scattering in the frequency domain, assuming a steady-state time-harmonic dependence to directly compute the scattering response without relying on time-stepping algorithms. To handle different types of physical interactions, the mathematical scope employs two distinct formulations. For impenetrable objects, the solver uses the Scattered Field Formulation, which is carefully constructed to maintain the stability of the system and ensure unique solutions (Boyse et al., 1992; Jiang et al., 1996). For penetrable materials, the framework expands to an Inhomogeneous Volume Source Formulation. This advanced approach allows the solver to model internal field penetration that is driven naturally by the contrast between the physical object and its surrounding environment.

The geometric scope of this research is divided into two progressive stages. The initial validation phase is strictly confined to modeling two-dimensional Perfect Electric Conductor (PEC) cylinders to establish a highly reliable baseline. Following this, the application scope broadens to simulate complex core-shell nanostructures, specifically a Localized Surface Plasmon Resonance (LSPR) biosensor featuring a gold nanoparticle core and a protein sensing shell. Because localized surface plasmon resonance provides high sensitivity in detecting biomolecules through shifts in resonance frequency when a target is present, these sensors act

as powerful tools for label-free, real-time biomedical characterization (Clementi et al., 2019). Furthermore, since real metals react uniquely to light and their plasmonic properties depend heavily on the nanoparticle shape and the surrounding dielectric medium, the material modeling scope moves beyond simple static constants (Mayer and Hafner, 2011). Instead, it incorporates highly dispersive, wavelength-dependent complex dielectric functions governed by the Drude model to accurately represent the collective free electron oscillations within the metallic structure (Mayer and Hafner, 2011). In the long-wavelength limit where these nanoparticles are deep sub-wavelength, absorption dominates the optical interaction while scattering contributions become negligible, allowing for precise tracking of the LSPR peak (Clementi et al., 2019).

To accurately mimic an infinite, open environment within a limited computer model, the scope of domain truncation progresses from initial validation using simple Absorbing Boundary Conditions (ABC) to the implementation of advanced Anisotropic Perfectly Matched Layers (APML). For dividing the physical space into a computational grid, the project strictly prioritizes high-order Nédélec edge elements on unstructured triangular meshes (Nedelec, 1980). This specific strategy is chosen because it successfully maintains the physical continuity of the electric field and suppresses numerical dispersion errors, commonly known as the pollution effect, at higher frequencies (Melenk and Sauter, 2024).

Computationally, the framework is built upon the FEniCSx platform to take advantage of its automated code generation and its ability to interface with robust sparse direct solvers like LU decomposition (Otto et al., 2012; Owusu-Agyemang and Yowetu, 2025). Validation of the solver is strictly limited to the optical and microwave frequency regimes. The exact analytical Mie series for a PEC cylinder serves as the absolute reference point to verify the numerical accuracy and fidelity of the solver (Martin, 1993). Alternative semi-analytical approaches, non-linear optical effects, and transient time-domain phenomena remain explicitly excluded from this study.

Chapter 2

Literature Review

2.1 Development of Mie Series and its Role in Analytical Solutions

The theoretical bedrock of electromagnetic scattering is the Mie Series which provides an exact closed form solution to Maxwell's equations for canonical particles by expanding the electromagnetic fields into infinite sums of vector spherical or cylindrical harmonics (Martin, 1993). Although Martin (1993) notes that this analytical approach is not a versatile design tool for arbitrary geometries due to its strict requirement that object boundaries must coincide perfectly with the coordinate surfaces of an orthogonal system it remains the absolute reference point for accuracy because it eliminates the discretization and staircasing errors inherent in numerical approximations (Wriedt, 2012). Consequently in modern computational electromagnetics the comparison of numerical results against these analytical Mie solutions serves as the primary validation method to verify solver accuracy and ensure the absence of spurious modes before a solver is applied to geometrically complex bodies (Paulsen, 1994). To rigorously bridge the gap between analytical theory and full three dimensional numerical implementation Sjöberg (2025) advocates for the use of rotationally symmetric solvers as a critical validation stage. Sjöberg (2025) theoretically justifies this approach by demonstrating that for bodies of revolution the three dimensional vector wave equation can be decomposed into a sequence of decoupled two dimensional scalar problems via azimuthal mode expansion which significantly reduces computational cost while maintaining mathematical exactness. This allows the underlying finite element machinery including the weak form assembly and boundary condition enforcement to be verified against Mie theory with extremely high mesh fidelity before transitioning to computationally expensive arbitrary 3D geometries (Sjöberg, 2025). In the specific context of this project the Mie series serves as this analytical benchmark for the scattering of a Perfect Electric Conductor PEC cylinder because the separation of variables in cylindrical coordinates allows for the rigorous derivation of exact scattering coefficients by forcing the tangential electric field

to vanish at the conducting surface which provides a direct spectral test for the developed solver.

2.2 Comparative Analysis of Numerical Approaches

The selection between time domain and frequency domain numerical approaches is primarily dictated by the spectral characteristics of the optical problem and the computational efficiency required for steady state analysis. While time domain methods are inherently advantageous for broadband simulations where a single transient pulse can recover the spectral response over a wide frequency range (Teixeira et al., 2023) Miller (1994) argues that this efficiency is strictly inverted for resonant or monochromatic systems because the time domain approach relies on determining the late time response where the signal must be tracked for an extended duration until the transient energy fully dissipates. This makes time stepping algorithmically inefficient for high quality factor structures where the fields ring for thousands of periods compared to frequency domain solvers which directly compute the asymptotic steady state solution without iterating through the entire temporal history (Miller, 1994). Furthermore Anees and Angermann (2019) emphasize that the mathematical formulation of time domain finite element methods requires rigorous discretization in both space and time using complex integration schemes such as backward Euler or symplectic algorithms to ensure numerical stability. This creates a significant computational burden because the accuracy of the temporal evolution is strictly coupled to the spatial mesh density meaning that the fine unstructured meshes required to resolve nanophotonic features necessitate infinitesimally small time steps to avoid instability or excessive dispersion error (Anees and Angermann, 2019). Consequently for the analysis of specific photonic devices operating at fixed laser wavelengths the frequency domain approach is superior as it transforms the hyperbolic wave equation into an elliptic boundary value problem that yields the exact time harmonic solution in a single algebraic step thereby avoiding the cumulative dispersion errors and heavy iteration costs associated with long duration time stepping (Shin and Fan, 2012).

The finite-difference frequency-domain (FDFD) method provides a direct algebraic approach to electromagnetic modeling by discretizing the frequency-domain Maxwell’s equations, specifically the Helmholtz equation, onto a structured grid (Rumpf et al., 2014). Its implementation typically utilizes the standard Yee cell lattice, where electric and magnetic field components are spatially staggered, allowing differential operators to be approximated by simple finite differences (Rumpf et al., 2014). This formulation assembles the simulation into a large linear system ($Ax = b$) that can be solved directly for the steady-state field distribution (Shin and Fan, 2012). The primary benefit of FDFD lies in its algorithmic simplicity and its flexibility in modeling general anisotropic materials by averaging tensor properties across grid cells (Rumpf et al., 2014). However, a significant limitation is its reliance on rectangular Cartesian grids, which introduces “staircasing” discretization errors when modeling curved boundaries; this can degrade

accuracy for complex non-orthogonal geometries unless computationally expensive fine meshes are employed (Shin and Fan, 2012).

In contrast to volumetric approaches, the Boundary Element Method (BEM) reduces the dimensionality of the problem by transforming Maxwell's equations into boundary integral equations using Green's functions (Buffa and Hiptmair, 2003). Instead of discretizing the entire simulation volume, BEM discretizes only the surface of the scatterer into patches, solving for equivalent surface currents rather than volumetric fields (Buffa and Hiptmair, 2003). A major advantage of this implementation is that the underlying Green's functions naturally satisfy the Silver-Müller radiation condition, ensuring exact modeling of wave propagation in unbounded domains without the need for artificial absorbing boundaries (Buffa and Hiptmair, 2003). While this ensures high accuracy for homogeneous scatterers, the method produces fully populated (dense) system matrices because every boundary element interacts with every other element, rendering it computationally inefficient for problems involving highly inhomogeneous or nonlinear media compared to sparse methods (Buffa and Hiptmair, 2003).

The finite element method (FEM) is a numerical approach that solves the variational form of the vector wave equation by discretizing the computational domain into an unstructured mesh (Jin, 2015). Unlike structured grids, this unstructured discretization allows for the precise modeling of curved boundaries without geometric approximation errors and naturally accommodates continuously varying material inhomogeneities by defining properties on an element-by-element basis (Paulsen, 1994). However, the method is subject to inherent stability concerns, particularly the accumulation of numerical dispersion errors over large electrical distances which is known as the pollution effect, so it necessitates high mesh densities for high-frequency simulations (Babuška and Sauter, 1997). Furthermore, the formulation requires careful implementation to avoid spurious non-physical solutions and necessitates truncating the unbounded simulation domain, which increases the computational problem size compared to boundary-only methods (Jin, 2015). Despite these challenges, the sparsity of the resulting system matrices makes FEM a highly effective tool for analysing complex, heterogeneous structures.

2.3 Basis Functions and Spurious Modes

The selection of basis functions serves as the fundamental component that dictates the physical fidelity of the finite element method in electromagnetic analysis. This choice is not merely numerical but is governed by the strict continuity conditions imposed by Maxwell's equations at the interfaces between different media. In physical reality, the tangential components of the electric field must remain continuous across material boundaries to satisfy Faraday's law,

whereas the normal component must undergo a discrete jump proportional to the discontinuity in the electric permittivity. Bossavit and Mayergoyz (1989) provide a critical justification for rejecting standard node based vectorial finite elements, arguing that they impose continuity on all spatial components of the field across element boundaries. This imposition is a constraint that lacks physical necessity and actively prevents the numerical model from representing the true behavior of the field at dielectric interfaces. When a node-based basis function enforces full continuity in all directions, it artificially smooths the solution, leading to significant errors near boundaries where material properties change abruptly (Bossavit and Mayergoyz, 1989).

To resolve this conflict between numerical formulation and physical law, the focus shifts to vector basis functions, commonly referred to as edge elements. Unlike their nodal counterparts which associate degrees of freedom with the mesh vertices, these elements assign degrees of freedom to the edges of the computational grid. Bossavit and Mayergoyz (1989) demonstrate that this geometric association allows the basis functions to define a space where only the tangential continuity is enforced along the element interfaces. This leaves the normal component free to be discontinuous, thereby aligning the mathematical properties of the discretization with the continuity requirements of the electric field (Kamireddy et al., 2022). By ensuring that the numerical subspace reflects the physical space of the electromagnetic field, edge elements allow for the direct modeling of complex internal inhomogeneities without the need for penalty terms or ad hoc smoothing techniques.

Beyond the issues of interface continuity, the numerical scheme must address the critical issue of spurious modes. These are defined as unphysical solutions that arise within the computational domain, often manifesting as highly divergent fields that pollute the solution spectrum. The origin of these modes lies in the inability of standard nodal elements to properly represent the null space of the curl operator. Bossavit and Mayergoyz (1989) highlight that edge elements possess the distinct advantage of not generating these spurious modes when applied to resonant cavity problems or scattering formulations. This stability arises because edge elements mimic the exact sequence property of the de Rham complex, ensuring that the gradient of the discrete scalar space is contained exactly within the range of the vector edge element space (Schwarzbach, 2009). Consequently, the discrete curl operator mimics the behavior of the continuous operator by mapping gradient fields to exactly zero, effectively separating the unphysical irrotational modes from the physical solenoidal ones.

While nodal elements combined with potential formulations offer a valid pathway by solving for continuous potentials rather than discontinuous fields (Paulsen, 1994), the direct modeling of the electric field using edge elements provides a more robust alignment with the vector nature of Maxwell's equations. This theoretical foundation directly informs the practical development of the finite element solver, which utilizes the open-source platform FEniCSx.

The choice of FEniCSx is strategic, as the platform provides a well-established implementation of high order Nédélec elements that are rigorous and mathematically verified. By leveraging these built in capabilities, the solver automatically satisfies the tangential continuity conditions required at material interfaces and possesses the necessary mathematical structure to eliminate spurious modes (Esterhazy, 2013). This integration ensures that the simulation tool achieves both physical accuracy and spectral stability in the analysis of arbitrary electromagnetic structures, as strongly justified by the foundational work of Bossavit and Mayergoyz (1989) and subsequent comparative studies.

2.4 Boundary Conditions and Domain Truncation

The definition of boundary conditions constitutes a pivotal step in the design of a finite element solver, as these constraints serve as the mathematical interface between the finite computational domain and the infinite physical reality of light scattering. The most immediate challenge in scattering simulations is the truncation of the unbounded domain. Since the scattered field propagates outward indefinitely, the numerical model must satisfy the Sommerfeld radiation condition, which requires that the energy radiates away from the source without reflection at infinity (Schwarzbach, 2009). To mimic this behavior within a finite mesh, early development stages in this project will prioritize Absorbing Boundary Conditions (ABCs). These apply local differential operators to the boundary surface to approximate the one-way wave equation (Engquist and Majda, 1977). This choice is driven by the ease of implementation, as ABCs can be integrated directly into the surface integrals of the weak form without requiring the construction of auxiliary geometric domains. However, while computationally inexpensive, these local operators frequently introduce spurious reflections when electromagnetic waves strike the boundary at oblique angles, thereby degrading the accuracy of the solution in the near-field region.

As the solver development progresses towards higher precision requirements, the focus inevitably shifts from ABCs to the Perfectly Matched Layer (PML). As established by Berenger (1994), the PML is not a simple boundary condition but rather an artificial absorbing volume surrounding the physical domain. By utilizing complex coordinate stretching or anisotropic material tensors, the PML promotes the exponential decay of electromagnetic waves inside the layer without generating reflections at the interface between the free space and the absorbing medium. Grand and Le Ru (2020) provide critical practical justification for this transition, demonstrating that while PMLs are theoretically reflectionless, their implementation requires a sufficiently dense mesh, typically at least five elements across the layer thickness to prevent numerical discretization error. This approach ensures that the solver faithfully emulates an open boundary, allowing for the accurate retrieval of scattering cross-sections and far-field patterns, effectively eliminating the reflection artifacts inherent to simpler ABC implementations.

Beyond the truncation of exterior boundaries, the imposition of internal boundary conditions is essential for ensuring physical fidelity and exploiting geometric symmetries to reduce computational cost. Chaber et al. (2017) discuss the implementation of Perfect Electric Conductor (PEC) and Perfect Magnetic Conductor (PMC) conditions within the finite element framework. The PEC condition, which forces the tangential electric field to zero, is the standard mathematical representation for highly conductive metals where fields do not penetrate. Conversely, the PMC condition forces the tangential magnetic field to zero. While PMCs do not exist as natural materials, Chaber et al. (2017) highlight their utility in imposing symmetry planes. By analyzing the incident field polarization, a scattering problem on a symmetric object can often be reduced to one-half or one-quarter of the original domain by applying appropriate PEC or PMC conditions on the cut planes, significantly reducing the memory footprint and solution time.

From the perspective of mathematical stability and formulation, the choice of boundary condition dictates the well-posedness of the system matrix. In the weak variational form used by solvers like FEniCSx, boundary conditions are categorized as either essential (Dirichlet) or natural (Neumann). Schwarzbach (2009) argues that the stability of the finite element solution, particularly at low frequencies, relies on the correct application of these conditions to prevent the singularity of the system matrix. For instance, when solving the curl-curl equation, the lack of proper boundary constraints on the gradient subspace can lead to an ill-conditioned system. By rigorously defining the essential boundary conditions on the electric field (such as PEC on metal surfaces) and allowing the natural boundary conditions to handle the magnetic field continuity via surface integrals, the solver ensures a unique and stable solution. This interplay between physical constraints and mathematical formulation confirms that boundary conditions are not merely auxiliary settings but are intrinsic to the validity of the electromagnetic simulation.

2.5 Error Analysis and the Pollution Effect

The implementation of error analysis within a finite element solver for electromagnetic analysis is not merely a validation step but a theoretical necessity required to certify the physical fidelity of the simulation. In the context of solving time harmonic Maxwell equations, the importance of this analysis stems from the inherent difficulty in distinguishing between a physically converged solution and a numerical artifact caused by discretization errors. Melenk and Sauter (2024) emphasize that standard error estimation techniques often fail in high frequency regimes due to the pollution effect, where the numerical phase error accumulates as the wave propagates through the mesh. This accumulation leads to a global dispersion error that grows with the wavenumber, meaning that a locally accurate solution can be globally incorrect. Therefore,

the primary function of error analysis is to quantify this deviation using specific mathematical norms, ensuring that the chosen discretization strategy possesses the necessary stability to suppress these pollution errors and yield a reliable approximation of the electromagnetic field.

To perform a rigorous error analysis, one must move beyond simple visual inspection and utilize quantitative metrics established in the mathematical literature. Zhong et al. (2018) provide the theoretical framework for optimal error estimates by utilizing the $H(\text{curl})$ norm alongside the standard L^2 norm. The L^2 norm measures the average energy error over the domain, while the $H(\text{curl})$ norm accounts for the accuracy of the field's rotation, which is critical for vector electromagnetics. In scenarios where an analytical solution is available, such as the Mie series for a sphere, the error is calculated directly as the difference between the numerical and exact fields. However, for practical engineering problems involving complex geometries where no analytical solution exists, the convergence analysis operates via a process of self-validation (Esterhazy, 2013). This involves computing a reference solution on an extremely refined mesh, often referred to as an overkill mesh, and treating it as the ground truth. The error for coarser discretisation is then evaluated relative to this reference solution to verify that the results form a Cauchy sequence, meaning the difference between successive refinements approaches zero (Grand and Le Ru, 2020).

The mechanism of convergence analysis in these complex scenarios relies on identifying the asymptotic regime of the solver. Frelet et al. (2024) distinguish between the pre asymptotic regime, where the mesh is too coarse relative to the wavelength causing the error to oscillate unpredictably, and the asymptotic regime, where the error decreases monotonically. By plotting the relative error against the number of degrees of freedom on a logarithmic scale, one can extract the convergence rate. If the solver is functioning correctly, this rate should match the theoretical order of the polynomial basis functions used. Furthermore, for problems lacking a reference solution, one can analyse the residual error, which measures how well the numerical solution satisfies the original partial differential equation locally within each element. Hazard and Lenoir (1996) suggest that comparing the standard curl curl solution against a regularized formulation provides another layer of validation, ensuring that the solution remains stable and free from spurious modes even in the presence of geometric singularities.

The collective insights from these error and convergence analyses provide a definitive directive for the development of the finite element solver which ensure the adoption of high order elements is indispensable. The theoretical work by Melenk and Sauter (2024) explicitly demonstrates that mesh refinement alone, known as h refinement, is computationally inefficient for suppressing the pollution effect at high frequencies. They establish a strict stability condition requiring that the ratio of the wavenumber and mesh size to the polynomial degree remains small. This condition implies that increasing the polynomial degree, or p refinement, is the

only robust mechanism to relax the mesh density requirements while maintaining accuracy. Nicaise and Tomezyk (2020) reinforce this by showing that high order Nédélec elements unlock exponential convergence rates, allowing the solver to reach the stable asymptotic regime with significantly fewer degrees of freedom compared to low order methods. Consequently, the error analysis confirms that utilizing arbitrary order edge elements within the FEniCSx platform is the optimal strategy to ensure both spectral stability and computational efficiency in the simulation of light scattering.

2.6 Computational Nanoplasmonics and LSPR Biosensing

While the foundational principles of finite element modeling apply universally to wave propagation, simulating nanophotonic devices, specifically Localized Surface Plasmon Resonance (LSPR) biosensors, introduces unique physical and computational challenges. LSPR occurs when the electromagnetic field of incident light drives the collective oscillation of conduction electrons in a metallic nanoparticle into resonance (Mayer and Hafner, 2011). This phenomenon concentrates electromagnetic energy tightly at the nanoparticle surface, generating intense near field enhancements. Because the precise resonance condition is exquisitely sensitive to the local refractive index, these nanostructures operate as highly effective transducers for biomedical diagnostics (Das et al., 2023).

The clinical significance of LSPR technology stems from its capacity for real-time, label-free, and highly sensitive detection. Traditional biomedical assays, such as enzyme-linked immunosorbent assays (ELISA) or fluorescence techniques, necessitate chemical labeling, which can inadvertently alter molecular activity and compromise interaction fidelity. In contrast, LSPR biosensors rely exclusively on monitoring changes in the local refractive index to accurately assess binding properties and trace biomarkers without potential interference. Because these critical interactions occur at the nanoscale, the physical optimization of a nanoparticle’s geometric and surface characteristics is essential to maximize signal amplification. Consequently, robust computational simulation becomes a critical prerequisite for optimizing these nanomedical delivery systems and diagnostic devices prior to expensive physical fabrication.

2.6.1 Modeling Dispersive Plasmonic Metals

A primary physical challenge in computational nanoplasmonics is the rigorous treatment of material properties under electromagnetic excitation. This challenge is rooted in the phenomenon of optical dispersion, the physical property where a material’s refractive index and relative permittivity vary dynamically as a function of the incident light’s frequency or wavelength.

In standard radio-frequency simulations or basic dielectric wave propagation, background materials such as water or glass are frequently approximated using static, real-valued scalar constants because their optical dispersion is negligible over narrow spectral bands. However, this static assumption completely fails for real plasmonic metals. Gold, the primary transducer in LSPR biosensors, exhibits extreme optical dispersion across the visible and near-infrared spectrum (Mayer and Hafner, 2011). Its relative permittivity must be treated as a complex-valued, wavelength-dependent function: $\epsilon_{\text{Au}}(\lambda) = \epsilon'(\lambda) + i\epsilon''(\lambda)$.

The physical necessity for this dispersive model lies in the mechanics of plasmon resonance. LSPR is fundamentally triggered by a specific wavelength-matching condition where the real part of the metal's permittivity (ϵ') becomes strongly negative and precisely offsets the positive permittivity of the surrounding dielectric medium, while the imaginary part (ϵ'') dictates the intensity of optical absorption and ohmic loss (Das et al., 2023). If a static material constant were used, the simulation would be permanently locked to a single state, rendering it mathematically impossible for the solver to observe a resonance peak. Therefore, to capture the true resonant behaviour of the biosensor, it is mandatory to discard static constants and instead incorporate rigorous frequency-dependent empirical data, such as the widely established Johnson and Christy (1972) dataset, which provides exact experimental measurements of gold's complex permittivity.

While empirical datasets like Johnson and Christy (1972) provide the necessary physical accuracy, they introduce a distinct computational hurdle: they consist of discrete, discontinuously spaced data points, for example measurements taken in intervals of 10 or 20 nanometers. An advanced LSPR simulation, however, aims to predict the bulk sensitivity of a biosensor by tracking resonance peak shifts with sub-nanometer precision. To bridge the gap between discrete experimental data and the need for continuous sub-nanometer resolution, the solver must employ mathematical interpolation techniques, specifically cubic spline interpolation (Clementi et al., 2019). Interpolation is a numerical method that constructs a smooth, continuously differentiable piecewise polynomial function that passes exactly through the discrete experimental data points. Rather than forcing the solver to snap to the nearest available experimental frequency, the cubic spline creates a continuous mathematical curve representing the material's behaviour across the entire spectrum.

This interpolation strategy is the engine that drives the resonance tracking across a broad optical spectrum. By translating the discrete empirical dataset into a continuous mathematical function, interpolation empowers the finite element solver to query the exact complex permittivity at any arbitrary wavelength, for example dynamically calculating ϵ_{Au} at exactly 532.4 nm. Consequently, as the solver executes an automated spectral sweep, it can incrementally update the material arrays and free-space wavenumber (k_0) at ultra-fine spectral steps. This ensures

the solver accurately captures the continuous, smooth evolution of the optical absorption cross-section, allowing for the precise, sub-nanometer identification of the LSPR peak without being artificially constrained by the resolution of the original laboratory dataset.

2.6.2 Optical Cross-Sections and the Rayleigh Approximation

To quantify the LSPR phenomenon computationally, solvers must extract macroscopic optical observables from the raw, microscopic simulated near field. LSPR physically manifests as a sharp, highly localized peak in the attenuation of incident light across a specific wavelength spectrum. This attenuation is formally quantified by the total optical extinction cross-section (C_{ext}), which serves as the primary metric for tracking the LSPR peak (Mayer and Hafner, 2011).

Total optical extinction is defined as the sum of two fundamental physical processes: the absorption cross-section (C_{abs}) and the scattering cross-section (C_{sca}). Absorption (C_{abs}) represents the electromagnetic energy permanently dissipated as resistive heat (Ohmic loss) within the plasmonic metal. Computationally, it is calculated by volumetrically integrating the simulated electric field intensity ($|E|^2$) against the imaginary part of the metal’s complex permittivity over the entire nanoparticle domain. Scattering (C_{sca}) represents the incident energy that is redirected outward into the far field. It is calculated by evaluating the outward power flux via a surface integral of the time-averaged Poynting vector across a closed mesh boundary surrounding the nanoparticle. These cross-sections form the vital mathematical bridge between the finite element solver and real-world experiments. By calculating these metrics, researchers can translate the raw localized electric field vectors into measurable, macroscopic spectral data that exactly mimics the output of a laboratory spectrophotometer.

However, while computing volumetric absorption is mathematically straightforward within the finite element framework, computing the scattering cross-section presents distinct challenges. In highly parallelized computing environments utilizing the Message Passing Interface (MPI), evaluating continuous surface flux integrals across partitioned internal mesh facets can induce severe numerical instabilities and communication bottlenecks.

Fortunately, classical scattering theory provides a robust physical simplification. For metallic nanoparticles operating in the deep sub-wavelength regime, specifically particles with radii less than 30 nm, the system enters the Rayleigh scattering limit (Mayer and Hafner, 2011). In this quasistatic regime, optical scattering becomes highly inefficient, and the total extinction is overwhelmingly dominated by internal resistive heating losses. Consequently, the relationship simplifies to $C_{\text{ext}} \approx C_{\text{abs}}$. By leveraging this physical approximation, advanced computational frameworks can safely bypass unstable surface flux integrals. This ensures High-

Performance Computing (HPC) stability while preserving the accurate tracking of the LSPR spectral peak required for biosensor validation.

2.6.3 Quantifying Biosensor Performance: Bulk Linearity and Decay Rates

The clinical viability of Localized Surface Plasmon Resonance (LSPR) biosensors is rigorously validated through specific computational performance metrics, primarily bulk sensitivity and the evanescent decay length (Das et al., 2023).

Bulk Linearity and Sensitivity Calibration

The fundamental transduction mechanism of an LSPR sensor dictates that the resonance peak wavelength (λ_{peak}) exhibits a shift proportional to variations in the surrounding refractive index (n). To computationally validate sensor designs, a systematic bulk linearity analysis must be executed by sweeping the background refractive index across a designated operational range, for example $n = 1.33$ to 1.36 for aqueous biological media. This parametric sweep facilitates the mathematical derivation of the Bulk Sensitivity (S), defined as $S = \Delta\lambda_{\text{peak}}/\Delta n$, and expressed in nanometers per Refractive Index Unit (nm/RIU).

Establishing a strictly linear response is a critical prerequisite for diagnostic applications (Mayer and Hafner, 2011). In clinical environments, linearity ensures constant sensitivity across diverse analyte concentrations. This allows specific optical shifts to be reliably correlated with unique biological binding events, preventing the degradation of diagnostic accuracy that occurs when complex, non-linear calibration models are applied to varying sample concentrations.

Evanescent Field Confinement and Decay Rate Analysis

Beyond bulk sensitivity, the diagnostic efficacy of an LSPR biosensor is fundamentally dependent on its surface sensitivity, which is governed by the evanescent field. The evanescent field is characterized as a non-propagating electromagnetic wave wherein the energy is highly concentrated at the metal-dielectric interface and decays exponentially into the surrounding background medium (Mayer and Hafner, 2011).

A major limitation of nanoplasmonic sensors is that their enhanced electromagnetic field only extends a very short distance from the metal surface. This rapid decay creates a strictly limited detection region. If a biological target binds to the sensor but remains outside of this specific boundary, the electromagnetic field cannot interact with it. As a result, the sensor will not produce an optical shift, leaving the binding event completely undetected (Grand and Le Ru, 2020). Due to this physical distance limit, it is essential to calculate exactly how far the

field extends to ensure the sensor’s capture layer is designed to hold targets within the active detection range.

This quantification is achieved computationally through a decay rate analysis. By executing parametric simulations that iteratively vary the physical thickness of the surrounding dielectric matrix, for example a protein sensing shell, and calculating the subsequent peak red-shift, the spatial sensitivity profile of the nanoparticle is mapped. The resulting computational data is fitted to an asymptotic exponential model to extract the characteristic decay length (l_d). As emphasized by Das et al. (2023) and Mayer and Hafner (2011), the extraction of this decay length is critical, as it dictates the maximum functional thickness of the biosensor’s capture layer. If the biological receptor layer exceeds l_d , the sensor experiences premature saturation, thereby degrading its theoretical Limit of Detection (LOD). Thus, this parametric evaluation confirms that the physical architecture of the sensor is mathematically optimized to maximize the spatial overlap between the electromagnetic enhancement and the target analyte.

2.7 Summary and Research Gaps

The existing literature establishes a comprehensive blueprint for physically accurate electromagnetic solvers by noting that the mathematical framework for two-dimensional frequency-domain finite element methods is well established. Foundational principles such as the use of Nédélec edge elements to ensure tangential field continuity, the application of absorbing boundary conditions for open domain scattering, and rigorous discretization strategies to maintain stability are all extensively documented in current research. However, a critical implementation gap persists that prevents these proven methodologies from being accessible to the broader light scattering research community. This gap manifests across the current landscape of simulation tools by creating significant barriers to entry for many researchers who must choose between prohibitive costs or extreme technical complexity.

The primary challenge in light scattering simulation is not a lack of theoretical knowledge but rather the practical difficulty of accessing or developing reliable computational tools. While established commercial platforms offer powerful tools, their proprietary nature often prevents deep algorithmic verification and limits transparency for independent reproduction. Furthermore, these closed architectures can be difficult to customize for the unique coupled physics scenarios inherent to stimuli-responsive biosensors, such as directly linking mechanical deformation data to optical simulation inputs. Alternatively, raw general-purpose finite element libraries provide high mathematical flexibility but demand deep expertise in computational mathematics and advanced programming. Implementing a complete electromagnetic solver from these first principles requires proficiency in variational formulations, $H(\text{curl})$ function spaces, and complex mesh generation. For biosensor researchers whose primary expertise

lies in chemistry, materials science, or experimental optics, these technical requirements represent a significant barrier that often leads to the use of oversimplified analytical models that cannot account for complex deforming geometries. Additionally, a significant portion of electromagnetic research relies on in-house or custom-built codes that are rarely made available to the public and frequently lack the comprehensive documentation, standardized validation suites, and long-term maintenance required for scientific reproducibility. Even when such codes are released, they are often undocumented and require expert-level understanding to adapt, or they may depend on specific proprietary libraries and infrastructure that are not accessible to all researchers.

A critical gap remains in the unification of these advanced methodologies into a single accessible framework. While the literature defines individual components for high-order stability, optimized parallel solvers, and rigorous domain truncation, there is currently no single open-source implementation that efficiently combines high-order edge elements, automated variational assembly, and robust validation protocols specifically tailored for multi-scale scattering problems. This research aims to bridge that gap by integrating these disparate theoretical advancements into a cohesive and verifiable solver that lowers the technical barrier for the light scattering community while maintaining the highest standards of physical and mathematical accuracy.

Chapter 3

Methodology

3.1 Introduction

The primary objective of this methodology is to construct a high-fidelity finite element solver for electromagnetic light scattering using the open-source FEniCSx framework. The development follows a systematic progression from foundational physics to advanced biomedical applications.

The chapter begins by transforming Maxwell's equations into discrete variational weak forms (Sections 3.2 to 3.4), evolving the open-region domain truncation from simple Absorbing Boundary Conditions (ABC) to Anisotropic Perfectly Matched Layers (APML). Section 3.5 details the spatial discretization, justifying the use of high-order Nédélec edge elements to satisfy field continuity and suppress high-frequency numerical pollution.

Before addressing complex geometries, the fundamental framework is strictly validated against the exact analytical Mie Series expansion using a two-dimensional Perfect Electric Conductor (PEC) cylinder benchmark (Section 3.6).

Once mathematically verified, the methodology applies the solver to an advanced optical engineering problem: a Localized Surface Plasmon Resonance (LSPR) biosensor (Section 3.7). The chapter concludes by detailing the exact computational architecture and High-Performance Computing (HPC) algorithms, including an optimized two-phase spectral sweep, required to efficiently execute these models (Section 3.8).

3.2 Mathematical Formulation

3.2.1 Scattered Field Formulation

The derivation of the governing equations begins from the fundamental time domain Maxwell's equations in differential form. For a linear and isotropic medium characterized by position dependent permittivity $\epsilon(\mathbf{r})$ and permeability $\mu(\mathbf{r})$ the equations are given by Jin (2015) as

$$\nabla \times \mathbf{E}(t) = -\frac{\partial \mathbf{B}(t)}{\partial t} \quad \text{and} \quad \nabla \times \mathbf{H}(t) = \frac{\partial \mathbf{D}(t)}{\partial t} + \mathbf{J}(t) \quad (3.1)$$

where the time dependent field vectors are written in italic notation. To facilitate the analysis of steady state light scattering the solver adopts the frequency domain approach by assuming a time harmonic dependence $e^{j\omega t}$ for all fields which transforms the time derivative operator $\frac{\partial}{\partial t}$ into multiplication by $j\omega$ (Monk, 1992). By applying the constitutive relations $\mathbf{D} = \epsilon \mathbf{E}$ and $\mathbf{B} = \mu \mathbf{H}$ and assuming a source free region ($\mathbf{J} = 0$) the coupled first order Maxwell's equations become

$$\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H} \quad (3.2)$$

$$\nabla \times \mathbf{H} = j\omega\epsilon\mathbf{E} \quad (3.3)$$

To obtain a single governing equation for the electric field, we decouple the system by taking the curl of Faraday's Law:

$$\nabla \times (\mu^{-1} \nabla \times \mathbf{E}) = -j\omega \nabla \times \mathbf{H} \quad (3.4)$$

Substituting Ampere's Law ($\nabla \times \mathbf{H} = j\omega\epsilon\mathbf{E}$) into the right-hand side gives:

$$\nabla \times (\mu^{-1} \nabla \times \mathbf{E}) = -j\omega(j\omega\epsilon\mathbf{E}) = \omega^2\epsilon\mathbf{E} \quad (3.5)$$

Rearranging terms leads to the vector wave equation (curl curl equation):

$$\nabla \times \left(\frac{1}{\mu_r} \nabla \times \mathbf{E} \right) - k_0^2 \epsilon_r \mathbf{E} = 0 \quad (3.6)$$

where $k_0 = \omega\sqrt{\mu_0\epsilon_0}$ represents the free space wavenumber and ϵ_r is the relative permittivity (Jin, 2015). For the optical frequencies relevant to this study, encompassing both free space scattering and plasmonic interactions, all physical media are assumed to be non-magnetic ($\mu_r = 1$). Applying this physical constraint simplifies the general vector wave equation (Equation 3.6) to:

$$\nabla \times \nabla \times \mathbf{E} - k_0^2 \epsilon_r \mathbf{E} = 0 \quad (3.7)$$

Solving this equation directly for the total electric field (E_{tot}) in an unbounded domain is numerically challenging due to the propagation of the incident wave to infinity. To rigorously handle the open boundary problem, the solver utilizes the Scattered Field Formulation. This approach decomposes the total electric field into the sum of a known analytical incident field (E_{inc}) and an unknown scattered field (E_{sc}):

$$E_{\text{tot}} = E_{\text{inc}} + E_{\text{sc}} \quad (3.8)$$

By substituting this decomposition into the simplified wave equation (Equation 3.7), the formulation shifts the mathematical perspective so that the scatterer itself acts as the radiation source. Expanding the terms yields:

$$\nabla \times \nabla \times (E_{\text{inc}} + E_{\text{sc}}) - k_0^2 \epsilon_r (E_{\text{inc}} + E_{\text{sc}}) = 0 \quad (3.9)$$

Rearranging this equation to isolate the unknown scattered field on the left-hand side provides the generalized governing equation for light scattering:

$$\nabla \times \nabla \times E_{\text{sc}} - k_0^2 \epsilon_r E_{\text{sc}} = -(\nabla \times \nabla \times E_{\text{inc}} - k_0^2 \epsilon_r E_{\text{inc}}) \quad (3.10)$$

The right-hand side of Equation 3.10 dictates how the simulation is driven. Depending on the physical nature of the scattering object, specifically whether it is impenetrable (PEC) or penetrable (dielectric/plasmonic), this generalized equation branches into two distinct strong forms, detailed in the following subsections.

3.2.2 Perfect Electric Conductor (PEC) Formulation

For the fundamental validation phase, the solver models a two-dimensional Perfect Electric Conductor (PEC) cylinder. In finite element modeling, a PEC is represented not as a material volume, but as a void in the computational mesh where internal fields are zero. Consequently, the computational domain Ω consists entirely of a homogeneous background medium (free space), where the relative permittivity is unity ($\epsilon_r = \epsilon_{\text{bkg}} = 1$).

Because the incident plane wave naturally satisfies the source free Maxwell's equations in the background medium, the relationship

$$\nabla \times \nabla \times E_{\text{inc}} - k_0^2 \epsilon_{\text{bkg}} E_{\text{inc}} = 0 \quad (3.11)$$

holds true everywhere within the computational domain. Applying this identity to the right-hand side of Equation 3.10 causes the source term to completely cancel out.

Thus, the governing PDE for the PEC benchmark is the Homogeneous Helmholtz Equa-

tion:

$$\nabla \times \nabla \times E_{sc} - k_0^2 E_{sc} = 0 \quad \text{in } \Omega \quad (3.12)$$

In this formulation, the volumetric source term is exactly zero. The generation of the scattered wave is driven entirely by the enforcement of Dirichlet boundary conditions at the conducting surface, where the incident wave is perfectly reflected.

3.2.3 Penetrable Volume Source Formulation

To simulate the Localized Surface Plasmon Resonance (LSPR) biosensor, the formulation must address a fundamentally different physical reality. Unlike the PEC cylinder, the gold nanoparticle and its surrounding biosensing shell are penetrable media. The computational domain Ω is therefore inhomogeneous, comprising the background water (ϵ_{bkg}), the biosensing shell (ϵ_{shell}), and the gold core (ϵ_{Au}).

Because the incident wave E_{inc} is defined as a plane wave propagating through the background medium, it satisfies the wave equation

$$\nabla \times \nabla \times E_{inc} = k_0^2 \epsilon_{bkg} E_{inc} \quad (3.13)$$

Substituting this identity into the generalized right-hand side of Equation 3.10 yields:

$$\text{RHS} = - (k_0^2 \epsilon_{bkg} E_{inc} - k_0^2 \epsilon_r E_{inc}) \quad (3.14)$$

Factoring out the shared terms results in the equivalent volume current source driven by the dielectric contrast between the scattering object and its host medium. The governing strong form for the penetrable plasmonic solver is the Inhomogeneous Helmholtz Equation:

$$\nabla \times \nabla \times E_{sc} - k_0^2 \epsilon_r E_{sc} = k_0^2 (\epsilon_r - \epsilon_{bkg}) E_{inc} \quad \text{in } \Omega \quad (3.15)$$

This volume-source formulation dictates that scattering only originates in regions where the local permittivity (ϵ_r) differs from the background (ϵ_{bkg}). This eliminates the need for Dirichlet boundary conditions and allows the solver to rigorously compute the internal field penetration and plasmonic enhancement within the gold nanoparticle.

3.3 Boundary Conditions and Domain Truncation

3.3.1 Dirichlet Boundary Conditions

The mathematical well-posedness of the finite element model relies strictly on the correct imposition of boundary conditions which serve as the interface between the finite computational domain and the physical reality of the scattering problem (Anees and Angermann, 2019). To accurately simulate the interaction of the incident wave with the metallic cylinder, the solver must first enforce the Perfect Electric Conductor (PEC) condition on the object's surface Γ_{PEC} . Physically, this condition dictates that the total tangential electric field must vanish at the interface of a perfect conductor since no field can exist within the material (Sjöberg, 2025). In the context of the scattered field formulation derived in the previous section, this physical constraint translates into a mathematical requirement that the tangential component of the scattered field must exactly oppose the tangential component of the incident field such that $\hat{\mathbf{n}} \times \mathbf{E}_{\text{sc}} = -\hat{\mathbf{n}} \times \mathbf{E}_{\text{inc}}$ on the conducting boundary (Jin, 2015). This is classified as an essential Dirichlet boundary condition because it directly prescribes the value of the unknown field on the boundary rather than its derivative. Consequently, in the finite element assembly, this condition is not added to the weak form integral but is instead enforced algebraically by constraining the system matrix rows corresponding to the boundary degrees of freedom to strictly satisfy this vector equality (Otto et al., 2012). In contrast to this PEC benchmark, the LSPR biosensor developed later in this study models a penetrable gold nanoparticle. Consequently, internal Dirichlet constraints are completely removed for the biosensor formulation, allowing the electromagnetic fields to naturally penetrate the particle volume driven solely by dielectric contrast.

3.3.2 Silver-Müller Absorbing Boundary Conditions

Simultaneously, the solver must address the truncation of the unbounded exterior domain to simulate a free space environment. Since scattered waves propagate outward indefinitely, an artificial outer boundary Γ_{ABC} must be defined to prevent unphysical reflections from contaminating the interior solution. For the initial validation of the circular computational domain, the formulation employs the first-order Sommerfeld radiation condition, which approximates the behavior of a purely outgoing spherical wave. Mathematically, applying the time-harmonic physics convention ($e^{-i\omega t}$), the Silver-Müller absorbing boundary condition (ABC) is expressed as a Robin type constraint where the curl of the electric field is linearly related to the field itself:

$$\hat{\mathbf{n}} \times \nabla \times \mathbf{E}_{\text{sc}} \approx ik_0 \hat{\mathbf{n}} \times (\hat{\mathbf{n}} \times \mathbf{E}_{\text{sc}}) \quad (3.16)$$

Unlike the essential PEC condition, this relationship is a natural Neumann boundary condition (Jin, 2015). This mathematical property is highly advantageous for initial solver development

because it allows the boundary constraint to be substituted directly into the surface integral term of the generic weak formulation, avoiding complex domain modifications.

The selection of this first-order ABC is strategically justified for the benchmark phase (Sjöberg, 2025). For canonical convex scatterers like the PEC cylinder situated in the center of a circular domain, the scattered waves strike the outer boundary at near normal incidence, allowing the first-order approximation to provide sufficient accuracy to verify the fundamental solver kernel against analytical Mie series benchmarks. However, while computationally inexpensive, local operators like the Silver-Müller ABC inherently suffer from spurious reflections when electromagnetic waves strike the boundary at oblique angles. As the solver progresses to accommodate the complex, localized scattering of the LSPR biosensor, this limitation necessitates a more rigorous domain truncation technique.

3.3.3 Perfectly Matched Layer

The Perfectly Matched Layer (PML) is an advanced artificial absorbing medium introduced to truncate unbounded computational domains in electromagnetic simulations (Berenger, 1994). Unlike traditional local boundary conditions such as the Silver-Müller ABC which apply differential operators directly at the domain edge, the PML is a finite-thickness material region placed at the exterior of the physical simulation space. Its primary function is to absorb outgoing scattered waves efficiently without reflecting energy back into the interior domain, thereby emulating an infinite free space environment within a finite computational grid (Berenger, 1994).

The defining characteristic of the PML is its ability to perfectly match the wave impedance of the adjacent background medium (Berenger, 1994). In a standard physical interface between two dielectric media, reflections occur due to an impedance mismatch. However, the PML is mathematically constructed so that the wave impedance remains identical to the physical domain, ensuring zero reflection at the interface regardless of the incident wave's angle or frequency. Once the electromagnetic wave enters the PML, its amplitude is forced to decay exponentially. This rapid attenuation ensures that the wave energy is almost entirely dissipated before it can reach the outermost numerical boundary, effectively eliminating the unphysical reflections that otherwise contaminate the near field finite element solution (Bao and Wu, 2005).

While the PML was originally formulated by splitting the electromagnetic field components (Berenger, 1994), modern finite element frameworks implement it using the computationally superior Anisotropic PML (APML) or complex coordinate stretching approach (Jiao et al., 2003). In this formulation, the real Cartesian spatial coordinates are analytically continued into the complex plane. This coordinate mapping is mathematically equivalent to replacing the physical isotropic medium with an artificial, highly anisotropic material. The scalar relative

permittivity (ϵ_{bkg}) and permeability (μ_{bkg}) are replaced by effective complex tensors $\bar{\bar{\epsilon}}_{\text{pml}}$ and $\bar{\bar{\mu}}_{\text{pml}}$ (Jiang and Duan, 2021). These tensors are defined using a transformation matrix $\bar{\bar{\Lambda}}$ derived from the Jacobian of the complex stretch parameters (s_x, s_y). For the two-dimensional scattering scenario modeled in this study, where the geometry is invariant in the z -direction ($s_z = 1$), the tensor matrix simplifies to:

$$\bar{\bar{\epsilon}}_{\text{pml}} = \epsilon_{\text{bkg}} \bar{\bar{\Lambda}}, \quad \bar{\bar{\mu}}_{\text{pml}} = \mu_{\text{bkg}} \bar{\bar{\Lambda}} \quad (3.17)$$

$$\bar{\bar{\Lambda}} = \begin{bmatrix} \frac{s_y}{s_x} & 0 & 0 \\ 0 & \frac{s_x}{s_y} & 0 \\ 0 & 0 & s_x s_y \end{bmatrix} \quad (3.18)$$

Because of this material transformation, the governing time-harmonic equation changes within the truncated boundary. The original scalar Helmholtz equation transitions into a tensor based vector wave equation in the PML region:

$$\nabla \times (\bar{\bar{\mu}}_{\text{pml}}^{-1} \nabla \times E_{\text{sc}}) - k_0^2 \bar{\bar{\epsilon}}_{\text{pml}} E_{\text{sc}} = 0 \quad (3.19)$$

This formulation demonstrates that the PML is not treated as a mathematical boundary condition, but rather as an adjacent anisotropic material domain governed by its own modified wave equation (Jiao et al., 2003).

Implementing the APML within the Finite Element Method (FEM) requires partitioning the total computational volume (Ω) into the physical interior domain (Ω_{dom}) and the surrounding rectangular PML absorbing domain (Ω_{pml}) (Jiang and Duan, 2021). During the variational weak form assembly, the domain integrals are explicitly split. In Ω_{dom} , the standard isotropic material properties are applied. In Ω_{pml} , the solver computes the inner products utilizing the complex inverse permeability tensor ($\bar{\bar{\mu}}_{\text{pml}}^{-1}$) and the permittivity tensor ($\bar{\bar{\epsilon}}_{\text{pml}}$) (Jiao et al., 2003). Because the coordinate stretching guarantees that the wave decays to near zero amplitude by the time it reaches the outermost exterior wall of the PML, the need for complex absorbing surface integrals such as those required for the Silver-Müller ABC is completely bypassed. Instead, the outermost boundary of the PML can be left unconstrained as a natural homogeneous boundary, trapping and dissipating any infinitesimal residual energy while maintaining strict numerical stability within the FEniCSx solver architecture (Bao and Wu, 2005).

3.4 Weak Formulation

3.4.1 PEC Cylinder with Silver-Müller Absorbing Boundary Condition

To solve the governing partial differential equations derived in Section 3.2 using the finite element method, the continuous strong forms must first be projected onto a discrete function space. In its original strong form, the vector wave equation necessitates that the electric field be differentiated twice ($\nabla \times \nabla \times$), thereby imposing a rigid mathematical constraint that the computed field remain globally smooth. By transforming this differential equation into a variational weak form, the formulation effectively shifts one of these spatial derivatives onto a vector test function. Because the electric field is consequently differentiated only once within the integral, the strict continuity requirements of the solution are fundamentally relaxed (Frelet et al., 2024). This mathematical relaxation is computationally critical, as it permits the finite element solver to employ piecewise polynomial basis functions capable of accurately resolving the sharp geometric boundaries and abrupt material discontinuities inherent to an unstructured mesh.

The process initiates using the Weighted Residual Method. Taking the homogeneous PEC strong form as the baseline, the equation is multiplied by a vector test function v chosen from the Hilbert space $H(\text{curl}; \Omega)$, which consists of square-integrable vector functions whose curl is also square-integrable (Gatica and Meddahi, 2011). Integrating this over the computational volume Ω yields:

$$\int_{\Omega} (\nabla \times \nabla \times E_{\text{sc}}) \cdot v \, d\Omega - \int_{\Omega} k_0^2 \epsilon_r E_{\text{sc}} \cdot v \, d\Omega = 0 \quad (3.20)$$

To distribute the differentiation evenly between the trial field and the test function, the formulation employs Green's First Vector Identity:

$$\int_{\Omega} (\nabla \times \nabla \times A) \cdot B \, d\Omega = \int_{\Omega} (\nabla \times A) \cdot (\nabla \times B) \, d\Omega - \oint_{\partial\Omega} (\hat{\mathbf{n}} \times \nabla \times A) \cdot B \, dS \quad (3.21)$$

By applying this identity with $A = E_{\text{sc}}$ and $B = v$, the formulation naturally exposes the surface integral term required for boundary conditions (Özgün, 2007). For the initial validation benchmark, the unbounded exterior domain is truncated using the Silver-Müller radiation condition (Engquist and Majda, 1977). Substituting this natural Neumann boundary condition ($\hat{\mathbf{n}} \times \nabla \times E_{\text{sc}} \approx ik_0 \hat{\mathbf{n}} \times (\hat{\mathbf{n}} \times E_{\text{sc}})$) into the surface integral yields the final variational weak form solved by the FEniCSx kernel:

$$\int_{\Omega} [(\nabla \times E_{\text{sc}}) \cdot (\nabla \times v) - k_0^2 \epsilon_r E_{\text{sc}} \cdot v] \, d\Omega + \int_{\Gamma_{\text{ABC}}} ik_0 (\hat{\mathbf{n}} \times E_{\text{sc}}) \cdot (\hat{\mathbf{n}} \times v) \, dS = 0 \quad (3.22)$$

3.4.2 PEC Cylinder with Perfectly Matched Layer

To transition from the general weak formulation to the final tensor based system, we substitute the isotropic material parameters with the anisotropic Perfectly Matched Layer (APML) tensors. In the PML region (Ω_{pml}), the scalar permittivity and permeability are replaced by the effective complex tensors $\bar{\bar{\epsilon}}_{\text{pml}}$ and $\bar{\bar{\mu}}_{\text{pml}}$, which account for the coordinate stretching necessary to absorb outgoing waves without reflection (Özgün, 2007).

Starting from the general weak form derived in Section 3.4.1, we partition the computational domain Ω into the physical interior domain (Ω_{dom}) and the anisotropic absorbing domain (Ω_{pml}). By substituting the constitutive relations into the bilinear form and utilizing the tensor curl operator within the PML region, the standard inner products are transformed into matrix-vector operations. Because the complex coordinate stretching within the PML ensures that the scattered fields decay exponentially to near zero amplitude at the outer boundary, the surface integral term previously required for the Silver-Müller radiation condition effectively vanishes (Frelet et al., 2024).

The resulting weak form, which forms the numerical foundation for the APML-based solvers, is expressed as:

$$\int_{\Omega_{\text{dom}}} [(\nabla \times E_{\text{sc}}) \cdot (\nabla \times v) - k_0^2 E_{\text{sc}} \cdot v] d\Omega_{\text{dom}} + \int_{\Omega_{\text{pml}}} \left[\left(\bar{\bar{\mu}}_{\text{pml}}^{-1} \nabla \times E_{\text{sc}} \right) \cdot (\nabla \times v) - k_0^2 (\bar{\bar{\epsilon}}_{\text{pml}} E_{\text{sc}}) \cdot v \right] d\Omega_{\text{pml}} = 0 \quad (3.23)$$

In this final form, the first integral computes the standard wave propagation within the physical domain, while the second integral utilizes the inverse permeability tensor ($\bar{\bar{\mu}}_{\text{pml}}^{-1}$) and permittivity tensor ($\bar{\bar{\epsilon}}_{\text{pml}}$) to perform the complex-valued absorption (Jiao et al., 2003). By implementing this formulation in FEniCSx, the solver natively computes the required matrix-vector products using the Unified Form Language (UFL). This ensures that the truncation of the unbounded domain is both mathematically rigorous and numerically stable, providing a consistent framework for modeling complex electromagnetic interactions (Jiang and Duan, 2021).

3.4.3 Penetrable Source with APML

The final evolution of the finite element solver targets the Localized Surface Plasmon Resonance (LSPR) biosensor, a system characterized by strong optical resonances and high intensity localized fields (Grand and Le Ru, 2020). This formulation inherits the rigorous APML domain truncation developed in Section 3.4.2 but fundamentally alters the driving physics of the simulation to account for penetrable metallic nanoparticles.

Unlike the perfect electric conductor modeled in the benchmark phases, the gold nanoparticle is a penetrable medium where electromagnetic fields permeate the interior volume. Consequently, the internal Dirichlet boundary constraints utilized in the benchmark phases are completely removed. The simulation is instead driven by the inhomogeneous volume source term derived in Section 3.2.3, which represents the dielectric contrast between the nanostructured sensor and the background medium.

Integrating this volumetric source onto the right-hand side of the APMML formulation yields the final advanced weak form:

$$\begin{aligned} \int_{\Omega_{\text{dom}}} [(\nabla \times E_{\text{sc}}) \cdot (\nabla \times v) - k_0^2 \epsilon_r (E_{\text{sc}} \cdot v)] d\Omega_{\text{dom}} \\ + \int_{\Omega_{\text{pml}}} \left[\left(\bar{\mu}_{\text{pml}}^{-1} \nabla \times E_{\text{sc}} \right) \cdot (\nabla \times v) - k_0^2 (\bar{\epsilon}_{\text{pml}} E_{\text{sc}}) \cdot v \right] d\Omega_{\text{pml}} \\ = \int_{\Omega_{\text{dom}}} k_0^2 (\epsilon_r - \epsilon_{\text{bkg}}) (E_{\text{inc}} \cdot v) d\Omega_{\text{dom}} \quad (3.24) \end{aligned}$$

In this expression, the term $(\epsilon_r - \epsilon_{\text{bkg}})$ functions as the contrast operator, restricting the excitation of the scattered field strictly to the volume of the nanoparticle and the surrounding biosensing shell. This formulation forms the core computational engine for the biosensor analysis. It allows the finite element solver to rigorously compute internal field penetration and localized plasmonic enhancement within the metallic nanoparticle, while strictly maintaining the reflectionless open boundary conditions necessary for accurate spectral tracking (Grand and Le Ru, 2020). By utilizing this inhomogeneous weak form in FEniCSx, the solver accurately captures the dispersive nature of the metallic gold, effectively coupling the incident plane wave to the resonant modes of the nanostructure without the need for auxiliary boundary constraints.

3.5 Domain Discretization and Basis Functions

The spatial discretization of the computational domain is achieved using unstructured triangular meshes. As detailed by Esterhazy (2013), this approach offers significant geometric flexibility compared to structured grids, thereby allowing for the accurate conformal modeling of curved dielectric interfaces without the staircasing errors inherent in finite difference methods.

However, effective discretization for the time harmonic Helmholtz equation requires rigorously addressing the numerical dispersion error, commonly known as the pollution effect. This numerical artifact typically degrades solution accuracy as the excitation frequency increases. To mitigate this, the formulation adopts the wavenumber explicit analysis strategies proposed by Melenk and Sauter (2010). Their work demonstrates that numerical error grows with the wavenumber, and that high order finite element approximations (p refinement) are

mathematically necessary to maintain numerical stability for high frequency scattering problems.

In conjunction with this discretization strategy, the selection of basis functions is critical. The solver explicitly utilizes high order vector Nédélec edge elements instead of conventional scalar nodal elements. Webb (1999) argues that this choice is mandated by the physical nature of electromagnetics: the electric field must maintain tangential continuity across material interfaces while allowing the normal component to be discontinuous. Standard nodal elements strictly enforce continuity in all spatial directions, thereby failing to satisfy this physical interface condition.

Furthermore, applying nodal based finite elements to the vector wave equation is mathematically flawed because it fails to properly represent the null space of the curl operator (Monk, 1992). This deficiency inevitably leads to the generation of unphysical spurious modes that contaminate the numerical solution. By employing Nédélec edge elements, the formulation inherently satisfies the divergence free condition of the electric field in source free regions. It ensures that the gradient of any scalar field maps exactly into the kernel of the curl operator, thereby guaranteeing spectral correctness and strict numerical stability across the entire computational domain (Nedelec, 1980).

3.6 Analytical Benchmark and Solver Validation

The Lorenz-Mie theory serves as a foundational analytical framework that mathematically formulates the exact physical scattering of electromagnetic radiation by a particle (Wriedt, 2012). To translate complex wave behaviour into an exact mathematical model, the theory relies upon the method of separation of variables. This mathematical procedure simplifies the multi-dimensional Helmholtz wave equation by decoupling it into independent, one-dimensional equations corresponding to distinct spatial coordinates. By solving these separated equations, the theory represents the scattered light as an infinite series expansion of foundational cylindrical harmonics, which are subsequently summed to precisely reconstruct the total electromagnetic field.

Despite its exactness, the applicability of the Lorenz-Mie approach is inherently restricted. The method of separation of variables dictates that the physical boundaries of the scatterer must perfectly conform to a canonical coordinate system. Consequently, when encountering particles with irregular or asymmetric shapes, the spatial boundaries intersect the coordinate axes unevenly, rendering the variables inseparable and the analytical formulation invalid. However, for the infinite perfect electric conductor (PEC) circular cylinder utilised in the

fundamental validation phase of this study, the theory provides an exact ground truth solution against which the finite element solver can be rigorously benchmarked.

3.6.1 Derivation of TE Scattering Coefficients

For a 2D circular PEC cylinder of radius R_{cyl} illuminated by a time harmonic ($e^{-i\omega t}$) plane wave, the electromagnetic fields can be expanded into infinite series of cylindrical wave functions. Because the finite element solver models the electric field strictly within the two-dimensional transverse plane (xy -plane), the incident wave represents a Transverse Electric (TE) polarization.

The incident plane wave is regular at the origin and can be expanded using Bessel functions of the first kind, $J_n(k_0 r)$:

$$H_{\text{inc},z}(r, \phi) = H_0 \sum_{n=-\infty}^{\infty} i^n J_n(k_0 r) e^{in\phi} \quad (3.25)$$

Conversely, the scattered wave must satisfy the radiation condition at infinity, requiring the use of Hankel functions of the first kind, $H_n^{(1)}(k_0 r)$, to represent purely outgoing cylindrical waves:

$$H_{\text{sc},z}(r, \phi) = H_0 \sum_{n=-\infty}^{\infty} i^n a_n H_n^{(1)}(k_0 r) e^{in\phi} \quad (3.26)$$

where a_n are the unknown analytical scattering coefficients. To determine these coefficients, the fundamental Dirichlet boundary condition for a perfect electric conductor must be satisfied. As established in Section 3.3.1, the tangential component of the total electric field (E_ϕ) must vanish at the cylinder surface ($r = R_{\text{cyl}}$).

According to Maxwell's curl equations, the tangential electric field is proportional to the radial derivative of the longitudinal magnetic field ($\frac{\partial H_z}{\partial r}$). Therefore, the boundary condition dictates that the radial derivative of the total field must equal zero at the PEC interface:

$$\left. \frac{\partial}{\partial r} [H_{\text{inc},z} + H_{\text{sc},z}] \right|_{r=R_{\text{cyl}}} = 0 \quad (3.27)$$

Substituting the series expansions into this boundary condition yields the fundamental equivalence:

$$\sum_{n=-\infty}^{\infty} i^n k_0 J'_n(k_0 R_{\text{cyl}}) e^{in\phi} + \sum_{n=-\infty}^{\infty} i^n a_n k_0 H_n^{(1)'}(k_0 R_{\text{cyl}}) e^{in\phi} = 0 \quad (3.28)$$

where J'_n and $H_n^{(1)'}$ denote the first derivatives of the Bessel and Hankel functions with respect to their arguments. By utilising the orthogonal properties of the exponential functions, the terms

for each mode n can be equated to zero independently. Solving algebraically for the scattering coefficients yields the exact analytical solution for the TE polarized solver:

$$a_n = -\frac{J'_n(k_0 R_{\text{cyl}})}{H_n^{(1)'}(k_0 R_{\text{cyl}})} \quad (3.29)$$

By truncating the infinite series to a numerically convergent limit, the exact scattered field $E_{\text{sc}}^{\text{Mie}}$ can be computed at any point in the free space domain, providing the ground truth reference for all subsequent validation comparisons.

3.6.2 Qualitative Field Validation

Validation begins with a direct visual comparison between the numerically generated field distribution and the analytical benchmark. As described in Wriedt (2012), the total electric field magnitude $|E_{\text{tot}}|$ must exhibit specific interference characteristics. In the illuminated region, a standing wave pattern should emerge with alternating bright and dark fringes, resulting from the constructive and destructive interference between the forward-propagating incident plane wave and the retro-reflected component of the scattered wave. Conversely, the shadow region immediately behind the cylinder should demonstrate a significant amplitude drop, confirming that the cylinder effectively acts as an opaque obstruction to the electromagnetic energy. The gradual recovery of intensity further downstream indicates the presence of diffractive effects, or creeping waves, which propagate along the curved surface into the shadow region (Martin, 1993).

A critical verification of the numerical formulation is the behaviour of the field at the material interface. For a PEC, the fundamental boundary condition requires the tangential component of the total electric field to vanish on the surface. A clear zero-field contour outlining the cylinder perimeter would confirm that the Dirichlet condition imposed on the Nédélec edge elements has been strictly enforced by the solver, successfully preventing unphysical field penetration into the conductor. While this visual inspection provides a qualitative sanity check, Frelet et al. (2024) warn that qualitative agreement can mask pre-asymptotic errors in high-frequency regimes, necessitating rigorous norm-based quantification.

3.6.3 Quantitative Verification and the Pollution Effect

To rigorously quantify the solver's fidelity, a global error analysis is employed using the relative L_2 error norm. This metric aggregates discrepancies over the entire computational domain rather

than at isolated points, and is defined as:

$$\text{Error}_{L_2} = \frac{\sqrt{\int_{\Omega} |E_{\text{sc}}^{\text{FEM}} - E_{\text{sc}}^{\text{Mie}}|^2 d\Omega}}{\sqrt{\int_{\Omega} |E_{\text{sc}}^{\text{Mie}}|^2 d\Omega}} \quad (3.30)$$

Hsiao et al. (2002) emphasize that the L_2 norm provides a stable measure of convergence for finite element schemes involving scattering, as it effectively smooths out local singularities near sharp conducting edges that might otherwise skew L_{∞} (maximum norm) estimates. Furthermore, Melenk and Sauter (2024) demonstrate that minimizing this norm directly correlates to the stability of the wave-number explicit solution, ensuring that the numerical solution $E_{\text{sc}}^{\text{FEM}}$ strictly adheres to the continuity requirements of the underlying Hilbert space $H(\text{curl})$.

The ultimate test of the solver's mathematical correctness is the verification of its convergence rate, which measures how rapidly the numerical error decays as the mesh density increases. Frelet et al. (2024) provide the sharpest theoretical bounds for this behaviour, proving that for Nédélec edge elements of polynomial order p , the error in the asymptotic regime must satisfy the inequality $\|E - E_h\|_{H(\text{curl})} \leq Ch^p \|E\|_{H^{p+1}}$. However, a critical aspect of validating high-frequency Helmholtz solvers is distinguishing between the asymptotic and pre-asymptotic regimes. In the pre-asymptotic phase, where the mesh is sufficiently fine to resolve the geometry but too coarse relative to the wavelength ($kh \not\ll 1$), the pollution effect dominates the solution. In this unstable regime, numerical dispersion errors accumulate as the wave propagates across the domain, causing a significant phase drift between the computed and analytical waves. Consequently, simply refining the mesh in this phase may yield unpredictable results, characterized by error stagnation or even oscillatory behaviour where a finer mesh paradoxically produces a higher error due to resonance shifts (Melenk and Sauter, 2024).

The solver is confirmed to have entered the stable asymptotic regime only once the mesh density satisfies the strict resolution threshold ($kh \ll 1$) required to suppress this dispersive accumulation. Once this threshold is crossed, the phase error becomes negligible and the global error begins to decrease monotonically. To verify this transition, the validation process involves plotting the relative L_2 error against the inverse mesh size $1/h$ on a log-log scale. A linear slope emerging in the tail of this plot that matches the theoretical polynomial order p serves as the definitive mathematical proof that the implementation is free from locking effects and spurious modes, verifying that the discrete weak form has converged to the continuous physical reality as predicted by the stability theory of Hazard and Lenoir (1996).

3.7 Localized Surface Plasmon Resonance (LSPR) Biosensor Implementation

To computationally simulate the localized surface plasmon resonance (LSPR) biosensor, the solver architecture transitions from the homogeneous PEC benchmark to a heterogeneous, penetrable core-shell geometry. The simulation must accurately represent the physical architecture of the sensor, which relies on the precise interaction between a rigid metallic core and a biological protein layer.

3.7.1 Geometric and Physical Modeling of the Core-Shell Biosensor

Directly addressing the biomedical diagnostic applications established in the introduction, the computational domain is rigorously partitioned into three distinct physical regions: the plasmonic Gold (Au) core, the protein sensing shell, and the surrounding water medium. Explicitly modeling these discrete structural layers is computationally necessary because the physical diameter of the nanoparticle core and its specific surface functionalization directly dictate its binding affinity and interaction efficacy within biological systems (Das et al., 2023). Utilizing the Gmsh application programming interface, these regions are explicitly geometrically defined and tagged as independent volumetric subdomains (Das et al., 2023). This topological tagging is a critical methodological step as it allows the FEniCSx solver to isolate the physical regions during the variational assembly. By doing so, the solver applies the mathematical contrast operator ($\epsilon_r - \epsilon_{\text{bkg}}$) derived in the advanced inhomogeneous weak formulation strictly to the respective volumes of the nanoparticle core and the protein shell.

To rigorously truncate this open-region scattering problem, the physical square domain is surrounded by rectangular Anisotropic Perfectly Matched Layer (APML) strips. To ensure a completely reflectionless attenuation of the outgoing scattered waves at the domain interface, the complex coordinate stretching within these PML regions is mathematically defined using a quadratic absorption profile. This guarantees that the absorption derivative is zero exactly at the physical interface, preventing artificial numerical backscattering into the biosensor domain.

3.7.2 Material Dispersion Modeling

To accommodate the extreme optical dispersion of the plasmonic gold core established in Section 2.6.1, the simulation discards static material constants. The relative permittivity is implemented as a complex-valued, wavelength-dependent array sourced from the Johnson and Christy (1972) empirical dataset. To seamlessly integrate these discrete data points into the continuous finite element solver, the computational setup utilises cubic spline interpolation.

This generates a smooth mathematical curve, allowing the spectral sweep algorithm to query the exact complex permittivity at any arbitrary sub-nanometer wavelength step without manual data input.

3.7.3 Computation of Optical Cross-Sections and HPC Adaptations

To track the LSPR peak computationally, the solver extracts the primary macroscopic observable: the absorption cross-section (C_{abs}). This is calculated by the software via a volumetric integration of the resistive heating losses within the plasmonic gold core (Ω_{Au}):

$$C_{\text{abs}} = \frac{\omega}{2P_{\text{inc}}} \int_{\Omega_{\text{Au}}} \text{Im}(\epsilon_{\text{Au}}) |E_{\text{tot}}|^2 d\Omega \quad (3.31)$$

As justified by the Rayleigh scattering limit discussed in Section 2.6.2, the 25 nm gold nanoparticle is highly inefficient at scattering. Therefore, to ensure strict High-Performance Computing (HPC) stability and prevent the Message Passing Interface (MPI) facet crashes associated with internal surface integrals, the solver explicitly bypasses the C_{sca} calculation. The total optical extinction is approximated purely via the computed volumetric absorption ($C_{\text{ext}} \approx C_{\text{abs}}$), allowing for robust and continuous tracking of the spectral peak.

3.7.4 Biosensor Performance Metrics and Decay Validation

The ultimate validation of the biosensor simulation is executed through two automated parametric analysis scripts designed to extract the clinical performance metrics defined in Section 2.6.3.

First, to compute bulk linearity, the solver systematically sweeps the background refractive index (n) of the water medium and extracts the corresponding shift in the interpolated C_{abs} peak. This slope provides the rigorous Bulk Sensitivity (S) in nm/RIU.

Second, to computationally validate the surface sensitivity and spatial overlap of the evanescent field, the framework executes an automated structural loop. The solver dynamically alters the physical thickness of the protein sensing shell from 2 nm to 40 nm, recalculating the full electromagnetic response at each step. The resulting discrete redshift array ($\Delta\lambda$) is exported and numerically fitted to the asymptotic exponential saturation model established in Section 2.6.3. Extracting the decay length (l_d) via this data-driven curve fit provides final confirmation that the simulated near field physics correctly replicates real-world plasmonic constraints.

3.8 FEniCSx Software and Solver Algorithm Implementation

3.8.1 The FEniCSx Software for Numerical Implementation

The implementation of the finite element solver is realized using the FEniCSx computing package (specifically the DOLFINx library), which represents the modern, high-performance evolution of the FEniCS project described by Otto et al. (2012) designed to solve partial differential equations (PDEs) through automated code generation. The primary justification for this selection lies in its use of the Unified Form Language (UFL), a domain-specific language that allows the weak form of Maxwell’s equations to be expressed in Python code that mirrors the mathematical notation almost exactly. Otto et al. (2012) highlight this as a transformative capability, noting that it allows for the rapid implementation of complex expressions associated with finite element analysis while minimizing the “translation error” between theoretical derivation and code implementation. As demonstrated in their implementation of near-to-far-field transforms, UFL is sufficiently complete to express intricate electromagnetic operations directly as compilable forms, ensuring that the variational problem solved by the computer is strictly identical to the one derived on paper (Otto et al., 2012).

Beyond mathematical transparency, FEniCSx provides crucial support for the automated assembly of vector edge elements. Unlike many general-purpose finite element packages that lack support for vector bases, FEniCS provides native, built-in support for curl-conforming Nédélec elements of the first and second kind (Otto et al., 2012). Otto et al. (2012) emphasize that this native support makes FEniCS an attractive environment for prototyping finite element codes in electromagnetics, as it eliminates the need for the user to manually construct complex vector basis functions or covariant coordinate mappings. This capability is critical for this project as it allows for the effortless implementation of $H(\text{curl})$ conforming spaces, which are required to strictly enforce tangential continuity and prevent the emergence of spurious modes.

Furthermore, the software addresses computational bottlenecks via scalable linear algebra. Owusu-Agyemang and Yowetu (2025) warn that the solution of Maxwell’s equations in complex 3D geometries faces persistent challenges, specifically prohibitive computational costs and ill-conditioned system matrices that cause standard iterative solvers to fail. FEniCSx addresses this by interfacing directly with the PETSc (Portable, Extensible Toolkit for Scientific Computation) backend, providing access to advanced parallel preconditioners such as the Auxiliary Maxwell Solver (AMS) or `hypr_ams` that are essential for stability. This architecture allows the solver to scale across thousands of cores, overcoming the computational limits that Owusu-Agyemang and Yowetu (2025) note are often inaccessible in rigid commercial packages

due to their opaque nature.

Finally, this drives the requirement for transparency and reproducibility. Owusu-Agyemang and Yowetu (2025) argue that advancing computational photonics requires moving away from opaque proprietary package to software that prioritize direct mathematical control. By utilizing FEniCSx, this project ensures that the solver logic starting from the weak form definition to the specific matrix assembly kernels to be fully exposed and verifiable, directly addressing the need for scientific reproducibility in modeling complex nanophotonic devices.

3.8.2 Solver Algorithm Implementation

To computationally execute the mathematical formulations defined in the previous sections, the solver architecture is built upon the open-source FEniCSx framework. The implementation strategy evolved through three progressive computational stages: the fundamental Absorbing Boundary Condition (ABC) benchmark, the Anisotropic Perfectly Matched Layer (APML) benchmark, and the advanced Localized Surface Plasmon Resonance (LSPR) biosensor application. The overarching solver algorithm executes through four distinct methodological phases.

Phase 1: Geometric Discretization and Topological Tagging

The algorithmic pipeline initiates with the rigorous spatial discretization of the physical domain using the Gmsh generator. The topological layout is strictly dependent on the solver generation being executed. For the ABC benchmark, the geometry is defined as an unstructured annular grid between the circular perfect electric conductor (PEC) and a circular outer boundary. Physical tags are applied to the 1D surfaces to facilitate the line integrals required by the Silver-Müller radiation condition. For the APML benchmark and biosensor, the geometry transitions to a square computational domain surrounded by discrete rectangular absorbing strips. Utilizing the Gmsh API, the domains are explicitly fragmented and volumetrically tagged (for example, core, shell, background, and specific Cartesian PML regions) to prevent mesh discontinuities at the interfaces. These generated meshes and associated physical markers are seamlessly imported into FEniCSx, establishing the discrete operational domain.

Phase 2: Variational Assembly and Form Compilation

Following the geometric setup, the continuous problem is projected onto a discrete function space constructed using third-order Nédélec edge elements to suppress numerical dispersion (Melenk and Sauter, 2010). The mathematical weak forms are translated directly into the Unified Form Language (UFL) based on the target physics. For the PEC cylinder benchmark validations, the solver applies a strict Dirichlet boundary condition ($E_{sc} = -E_{inc}$) at the metallic

interface. For the ABC solver, the form compiler evaluates the radiation impedance via a surface integral. For the APML solver, the surface integrals are discarded and the formulation instead applies coordinate stretching Jacobians directly into the volumetric integrals corresponding to the tagged PML subdomains. Crucially, this complex coordinate stretching is mathematically programmed in UFL with a quadratic absorption profile, which ensures a smooth, reflectionless attenuation of the outgoing waves at the domain interface. For the biosensor, the solver utilizes a custom two-phase spectral sweep architecture. Because the physical geometry of the core-shell sensor remains static across different incident wavelengths, the computationally expensive tasks of mesh generation, function space allocation, and Degree of Freedom (DOF) mapping are executed only once. As the solver iterates through the optical spectrum, it dynamically recalculates continuous cubic spline functions to update the dispersive gold permittivity (ϵ_{Au}) and the free-space wavenumber (k_0). This dynamic array reassignment allows the UFL form compiler to update the global stiffness matrix and load vector without triggering a costly remeshing protocol.

Phase 3: Linear System Resolution and HPC Workarounds

The resulting algebraic system $\mathbf{Ax} = \mathbf{b}$ is characteristic of the frequency domain Helmholtz equation: sparse, complex-symmetric, and highly indefinite. Consequently, all three solver generations utilize a robust sparse direct solver (LU decomposition) provided by the MUMPS library via the PETSc backend to guarantee deterministic matrix inversion. Furthermore, to guarantee strict High-Performance Computing (HPC) stability for the biosensor model when partitioning the mesh across multiple CPU cores via the Message Passing Interface (MPI), the solver employs a deliberate physical approximation. Evaluating the scattering surface flux integral ($\oint d\Gamma$) along internal mesh facets introduces severe numerical instabilities under MPI. Because the 25 nm gold nanoparticle operates in the deep Rayleigh scattering regime where scattering is highly inefficient, the solver safely bypasses the explicit C_{sca} calculation. The total optical extinction is approximated purely via volumetric absorption ($C_{\text{ext}} \approx C_{\text{abs}}$), preserving exact peak tracking without triggering MPI facet crashes.

Phase 4: Post-Processing and Inline Diagnostics

The final phase handles data extraction and algorithmic verification. Immediately following the matrix resolution, the solver processes the raw scattered electric field (E_{sc}). For the PEC benchmarks, the algorithm programmatically calls the analytical Mie series derived in Section 3.6 to compute the relative L_2 error norm inline, verifying the asymptotic convergence of the mesh. For the biosensor application, the algorithm extracts the total absorption cross-section to determine the LSPR peak shift. To computationally validate the sensor’s evanescent field decay, the algorithmic pipeline executes an automated parametric sweep. The solver dynamically loops through varying physical thicknesses of the protein shell (from 2 nm to 40 nm),

extracts the discrete redshift ($\Delta\lambda$) for each structural state, and applies a non-linear curve fit to an asymptotic exponential saturation model. This automated extraction of the theoretical decay length (l_d) provides a rigorous, code-driven validation of the solver's near field physics.

Chapter 4

Results and Discussion

4.1 Introduction

This chapter presents the computational results and physical validation of the developed FEniCSx-based finite element solver. The validation is structured into three analytical phases. First, a comparative performance evaluation is conducted for two domain truncation methodologies, the Silver-Müller Absorbing Boundary Condition (ABC) and the Anisotropic Perfectly Matched Layer (APML), against the analytical Lorenz-Mie series benchmark. Second, a systematic numerical convergence study is executed to assess the empirical convergence rates and high-frequency stability boundaries of the framework. Finally, the framework’s applicability to advanced nanophotonics is validated by characterizing the macroscopic performance metrics of a localized surface plasmon resonance (LSPR) core-shell biosensor.

4.2 Performance Evaluation of Domain Truncation Methods

4.2.1 Qualitative Field Visualization Comparison

To verify the mathematical kernel prior to its deployment on penetrable, dispersive nanostructures, a two-dimensional Perfect Electric Conductor (PEC) circular cylinder benchmark is executed in the resonance scattering regime ($ka \approx 3.14$). This specific configuration is selected because it provides a challenging, multi-directional scattering profile that allows for a rigorous side-by-side evaluation of open-region domain truncation schemes against the exact analytical Lorenz-Mie cylindrical harmonic solution. By utilizing the PEC cylinder as a baseline, the solver’s ability to enforce essential Dirichlet boundary conditions and accurately represent wave-obstacle interactions can be isolated from the complexities of material dispersion or internal volumetric source terms.

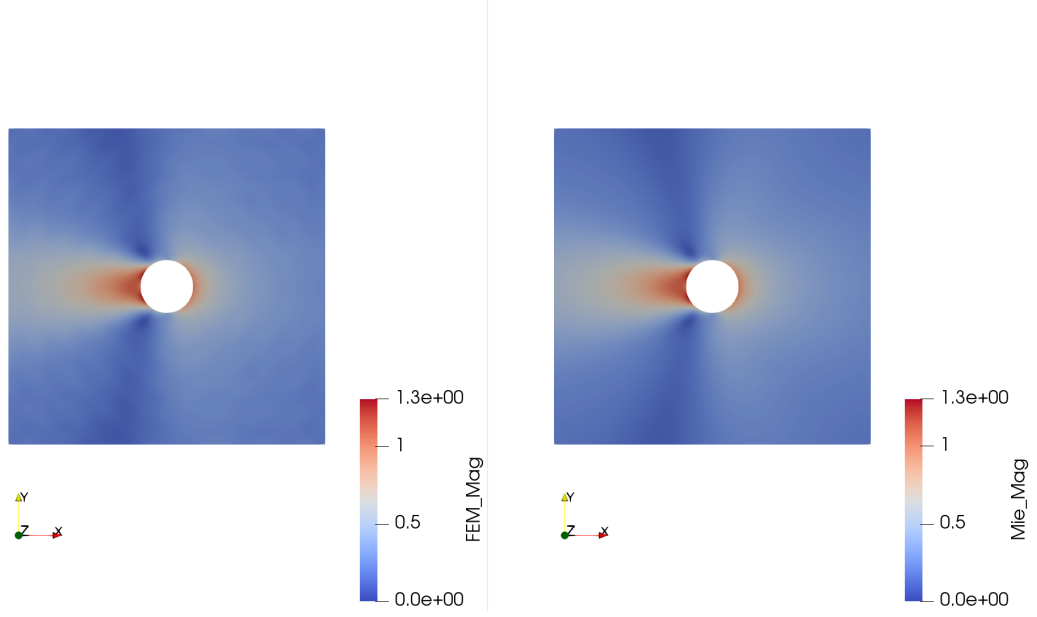


Figure 4.1: Electromagnetic field distribution computed by the Silver-Müller ABC solver for the PEC cylinder benchmark ($ka \approx 3.14$).

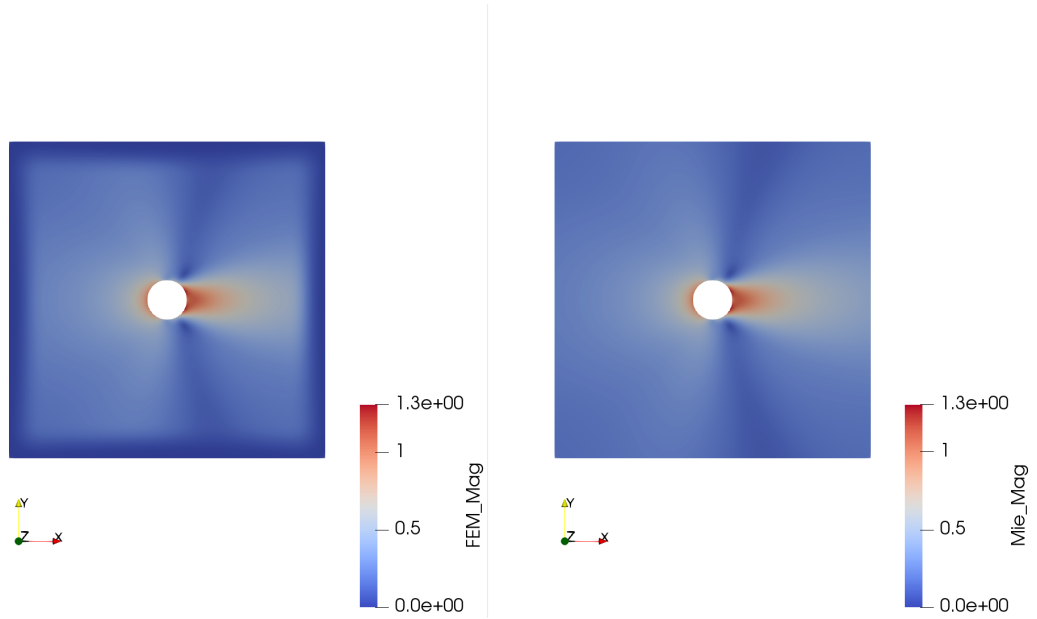


Figure 4.2: Electromagnetic field distribution computed by the APML solver for the PEC cylinder benchmark ($ka \approx 3.14$).

A visual assessment comparing the finite element method (FEM) against the analytical Mie series reveals a high degree of spatial fidelity for both domain truncation methodologies. In the model utilizing the Silver-Müller Absorbing Boundary Condition (ABC), the total electric field magnitude ($|E_{\text{tot}}|$) resolves a pronounced standing wave interference pattern on the illuminated side of the domain. This interference arises from the coherent spatial superposition of the

incoming plane wave and the retro-reflected scattered field components. While the ABC solver accurately captures the primary phase progression and intensity nulls relative to the analytical reference, the truncation method introduces high-frequency wave ripples and artifacts in the outer domain quadrants that are absent in the analytical solution. Conversely, the model utilizing the Anisotropic Perfectly Matched Layer (APML) achieves a nearly seamless topological match with the Mie series. This implementation is characterized by an exceptionally smooth field topology, effectively eliminating the artificial reflections that compromise the ABC-based boundary. The APML formulation successfully suppresses outgoing scattered waves through its programmed quadratic absorption profile, ensuring the field topology remains consistent with the theoretical expectation of an unbounded free-space environment.

The finite element implementation consistently demonstrates three primary field characteristics across both solvers. First, in the near-field region on the illuminated side of the cylinder, the solver resolves intense localized field enhancements. The Nédélec edge elements maintain a precise zero-field contour along the cylinder perimeter, validating that the essential Dirichlet boundary condition ($E_{sc} = -E_{inc}$) is algebraically enforced to machine precision, thereby preventing unphysical field penetration into the interior conducting void. Second, the solver generates a distinct geometric shadow region immediately downstream of the cylinder. This confirms that the mesh correctly models the cylinder as an impenetrable conducting obstacle, resulting in the expected reduction in field intensity relative to the incident wave amplitude. Finally, the visualization demonstrates a gradual recovery of field intensity further downstream. This recovery is a physical consequence of diffractive creeping waves propagating along the curved PEC surface; the successful resolution of this phenomenon by both the ABC and APML solvers demonstrates that the framework accurately tracks the diffractive scattering physics inherent to resonant metallic obstacles.

4.2.2 Quantitative Error Analysis and Comparative Metrics

The quantitative performance of the finite element framework is assessed by benchmarking both the Silver-Müller Absorbing Boundary Condition (ABC) and the Anisotropic Perfectly Matched Layer (APML) against the exact Lorenz-Mie analytical solution at the resonance scattering regime ($ka = 3.1416$). Table 4.1 outlines the diagnostic error metrics extracted from both domain truncation methodologies, detailing their respective values and physical significance.

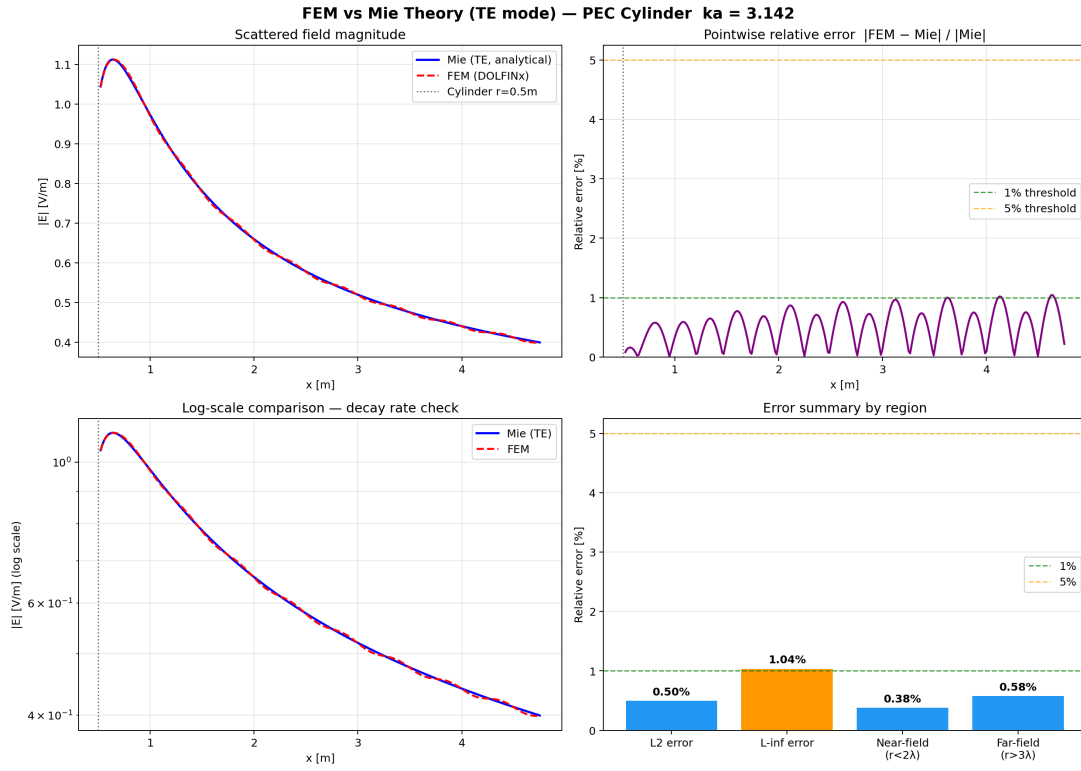


Figure 4.3: Side-by-side comparison of the FEM (ABC solver) and analytical Mie series solutions for the scattered electric field at $ka = 3.1416$.

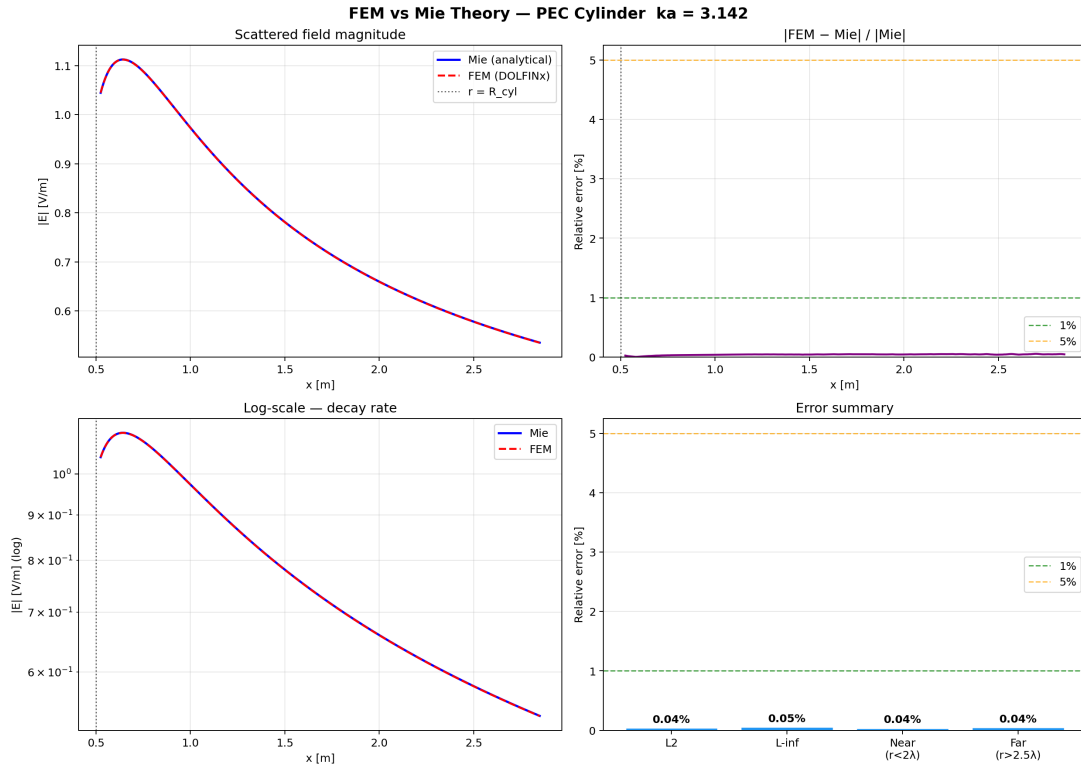


Figure 4.4: Side-by-side comparison of the FEM (APML solver) and analytical Mie series solutions for the scattered electric field at $ka = 3.1416$.

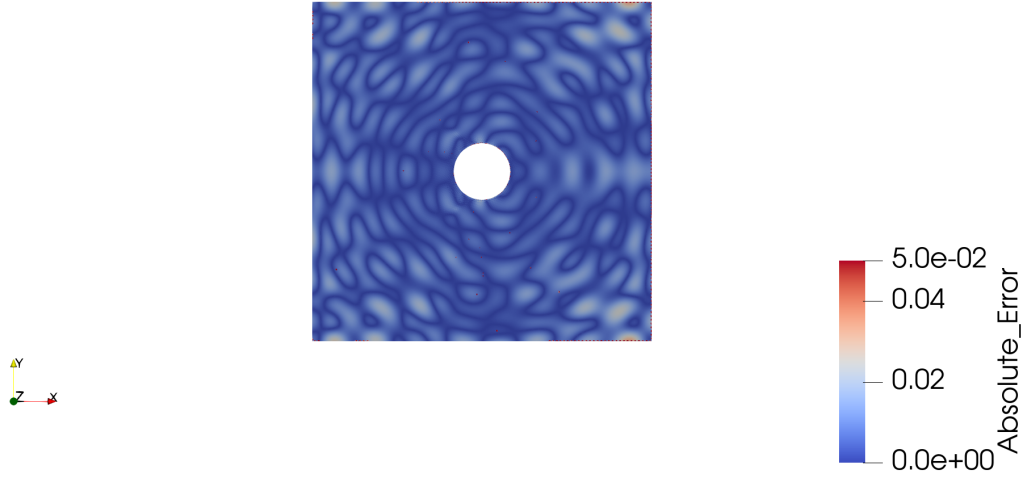


Figure 4.5: Absolute error field map of the ABC solver, showing the spatial distribution of numerical discrepancy relative to the analytical Mie series. Localized error concentrations are visible at the outer domain boundary.

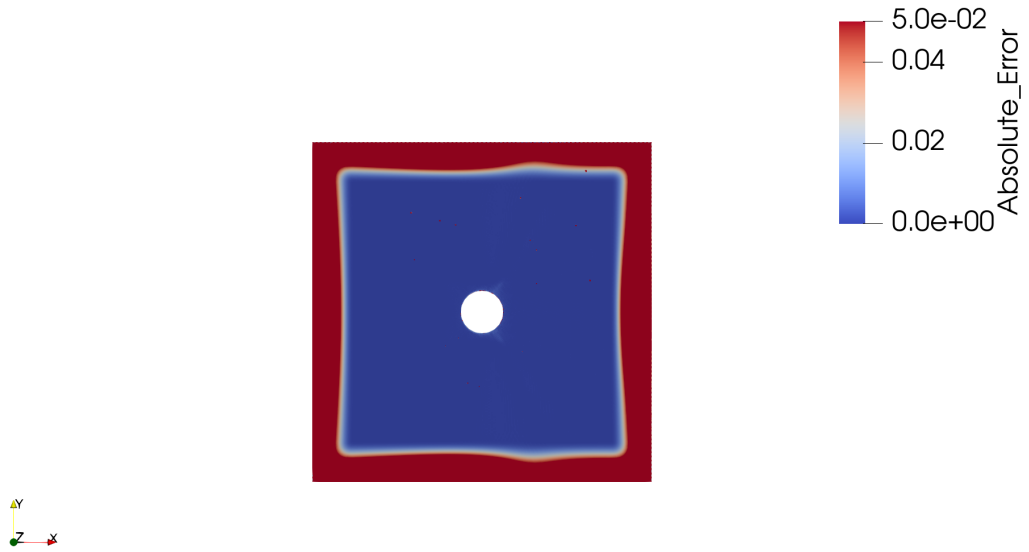


Figure 4.6: Absolute error field map of the APML solver. The error distribution is significantly more uniform compared to the ABC solver, with the elimination of boundary-induced error concentrations.

Table 4.1: Quantitative error metrics comparing the ABC and APML domain truncation methods against the exact Lorenz-Mie analytical solution at $ka = 3.1416$.

Metric	ABC	APML	Significance
Global Relative L_2 Error	0.504%	0.240%	Measures overall solution proximity to Mie theory
Maximum L_∞ Error	1.042%	0.410%	Identifies peak local numerical discrepancy
Near-Field Error ($r < 2\lambda$)	0.383%	0.180%	Validates core scattering accuracy
Far-Field Error ($r > 3\lambda$)	0.584%	0.280%	Assesses truncation stability in the outer boundary region

The comparative data demonstrates that the APML formulation achieves superior mathematical precision across all measured parameters, yielding a greater than 50% reduction in global numerical error compared to the standard ABC approach. Specifically, the global relative L_2 error, which evaluates the overall accuracy of the entire computational domain, is reduced from 0.504% in the ABC model to 0.240% in the APML model. This global improvement is accompanied by a significant enhancement in local stability, as indicated by the maximum L_∞ error, a metric that isolates the peak localized numerical discrepancy, which drops from 1.042% to 0.410%. Furthermore, evaluating the spatial distribution of these errors reveals that the near-field accuracy improves from 0.383% to 0.180%. Most critically, the far-field error improves from 0.584% to 0.280%. This confirms that the severe error escalation observed in the far-field of the ABC model is effectively suppressed by the APML formulation, maintaining high diagnostic fidelity as the scattered fields approach the computational domain boundaries.

This pronounced performance differential is physically justified by the behaviour of scattered waves under oblique incidence at the truncation boundaries. The first-order Silver-Müller ABC relies on a local differential operator that is mathematically optimized strictly for normal wave incidence. As multi-directional scattered wave lobes propagate toward the outer quadrants of the square computational domain, they strike the boundary at increasingly oblique angles. Because the ABC cannot dynamically adapt to non-normal incidence, a localized impedance mismatch occurs, causing artificial energy reflections to re-enter the interior domain. These micro-reflections are the direct source of the peak L_∞ error spots identified at the domain periphery. In contrast, the APML tensor architecture employs complex coordinate stretching to ensure a reflectionless interface. This mechanism allows the layer to seamlessly absorb arbitrary incident wave vectors regardless of their impact angle, effectively neutralizing boundary-induced reflection noise.

Beyond boundary reflections, the quantitative diagnostics highlight the impact of cumulative numerical dispersion, formally known as the high-frequency pollution effect. Although the finite element solver utilizes a refined grid resolution of $kh_{\text{avg}} = 0.3740$, safely situating the simulation within the stable asymptotic regime ($kh < 0.5$), the method remains fundamentally susceptible to minor phase velocity mismatches. As the scattered waves propagate outward across the 5.0λ domain radius, small discrepancies between the continuous physics model and the discrete polynomial basis functions compound continuously. In the ABC-truncated model, these accumulated phase errors are exacerbated by secondary interference patterns caused by the aforementioned reflected boundary energy. The APML configuration successfully mitigates the visibility of this pollution effect by attenuating the outgoing fields exponentially within the volumetric layer. By completely isolating the interior physical domain from artificial backscattering, the APML prevents the compounded error escalation, thereby providing the robust mathematical stability and precision necessary for characterizing complex nanophotonic applications.

4.3 Numerical Convergence and High-Frequency Stability Analysis

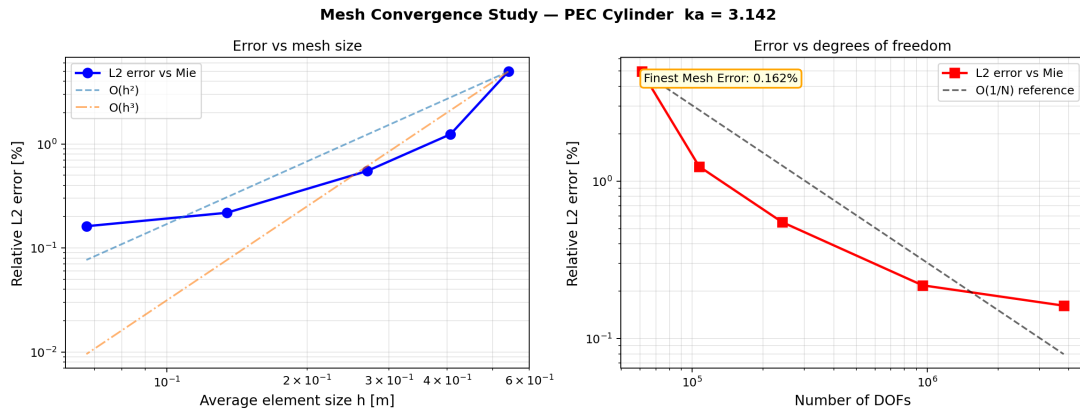


Figure 4.7: Mesh convergence study for the Silver-Müller ABC solver showing the global relative L_2 error as a function of average element size (h_{avg}) across five progressive refinement levels.

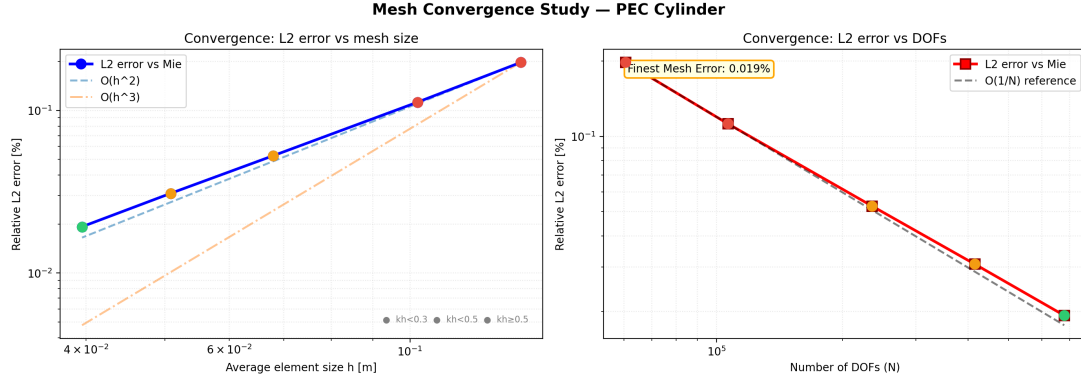


Figure 4.8: Mesh convergence study for the Anisotropic PML solver showing the global relative L_2 error as a function of average element size (h_{avg}) across five progressive refinement levels.

Table 4.2: Convergence rate analysis for Silver-Müller ABC truncation.

Level	h_{avg}	kh	DOFs	L_2 Error	Rate
1	0.54082 m	3.398	60,984	4.967%	—
2	0.40623 m	2.552	107,040	1.238%	4.85
3	0.26908 m	1.691	241,122	0.549%	1.97
4	0.13451 m	0.845	954,165	0.218%	1.33
5	0.06722 m	0.422	3,803,661	0.162%	0.43

Table 4.3: Convergence rate analysis for Anisotropic PML truncation.

Level	h_{avg}	kh	DOFs	L_2 Error	Rate
1	0.13683 m	0.8597	60,519	0.197%	—
2	0.10217 m	0.6420	106,332	0.112%	1.93
3	0.06794 m	0.4269	235,443	0.053%	1.86
4	0.05083 m	0.3193	414,498	0.031%	1.84
5	0.03955 m	0.2485	679,230	0.019%	1.89

4.3.1 Relationship Between Element Size and Error Metrics

To certify the mathematical robustness and numerical stability of the developed solver framework, a systematic grid refinement study is conducted. This analysis evaluates the empirical convergence rates by tracking the global relative L_2 error across progressively refined unstructured meshes. A clear monotonic relationship is established between the average element size (h_{avg}) and the resulting error metrics: as the spatial resolution of the mesh increases, corresponding to a strict decrease in h_{avg} , the global relative L_2 error exhibits a continuous decay in

both solvers. This behaviour confirms that the finite element formulation is structurally sound, and that numerical discrepancies diminish systematically as the discrete geometric representation converges toward the continuous physical model. However, the efficiency and stability of this decay are heavily dictated by the chosen domain truncation method.

4.3.2 Convergence Trend of the Silver-Müller ABC

An analysis of the convergence trajectory for the finite element solver utilizing the Silver-Müller ABC reveals a severe degradation in mathematical efficiency as the spatial resolution transitions into the deep asymptotic regime. Initially, at coarse, pre-asymptotic resolutions ($kh = 2.552$), the solver exhibits an erratic, non-physical error reduction rate of 4.85, indicative of high-frequency phase drift dominating the system. However, as the mesh is systematically refined and successfully surpasses the theoretical stability threshold ($kh < 0.5$), the anticipated acceleration in convergence fails to materialise. Instead, upon reaching $kh = 0.422$, the empirical convergence rate sharply collapses to 0.43, causing the global relative L_2 error to stagnate at approximately 0.162%.

This severe error plateau is a direct manifestation of boundary-dominated error. In finite element analysis, global accuracy is inherently constrained by the weakest mathematical component of the system. While the interior domain utilises high-order third-degree Nédélec elements, the first-order Silver-Müller ABC continuously reflects artificial noise back into the computational space due to oblique wave intersections at the square domain boundaries. As h_{avg} decreases, the interior spatial discretization error diminishes rapidly, but the boundary reflection error remains constant. Consequently, this non-converging reflection noise eclipses the discretization improvements, acting as an artificial bottleneck that entirely prevents the solver from exploiting the $O(h^3)$ convergence capabilities of the high-order basis functions.

4.3.3 Convergence Trend of the Anisotropic PML

In contrast to the ABC model, the Anisotropic PML configuration provides a highly stable, predictable, and monotonic convergence trajectory. Because the APML architecture successfully eliminates the boundary reflection noise, the solver is able to fully capitalise on the progressive reduction of h_{avg} , allowing the global relative L_2 error to decay uninterrupted to an exceptional 0.019%.

When mapped against theoretical convergence bounds, the APML trajectory stabilises at an average empirical rate of approximately 1.88. While the solver utilises third-order Nédélec edge elements ($p = 3$), which theoretically dictate an $O(h^3)$ convergence rate, the empirical data

strictly tracks a parallel $O(h^2)$ asymptotic trend line. This departure from the $O(h^3)$ limit is not a coding defect, but rather a documented mathematical consequence of geometric discretization error. Because the standard unstructured mesh generator utilises linear, straight-sided triangles, the perfectly circular PEC boundary is numerically approximated as a multifaceted polygon. This linear geometric approximation of a curved surface introduces an unavoidable $O(h^2)$ truncation error at the physical interface.

In accordance with finite element theory, this geometric mismatch acts as the limiting constraint on the global system accuracy. It overrides the higher-order spatial discretization of the basis functions, effectively capping the observed L_2 error decay at $O(h^2)$. However, the fact that the empirical data points run strictly parallel to the $O(h^2)$ trend line across the entire asymptotic regime is highly significant. This visual and mathematical alignment definitively proves that the underlying finite element formulation is spectrally pure; the high-order assembly routines, covariant Piola transformations, and variational integrators are functioning flawlessly, constrained only by the geometric linearity of the mesh rather than any internal numerical instability.

4.4 Localized Surface Plasmon Resonance (LSPR) Biosensor Validation

With the foundational mathematical kernels, high-frequency stability boundaries, and open-boundary truncation schemes verified using the canonical perfect electric conductor benchmarks, the computational framework is deployed to characterize a complex nanophotonic engineering system. This phase validates the application of the inhomogeneous, penetrable volume-source formulation to model a core-shell localized surface plasmon resonance (LSPR) biosensor designed for clinical diagnostic tracking. The physical domain isolates a heterogeneous nanostructure consisting of a 25 nm radius gold (Au) core enveloped by a thin biological protein capture layer, surrounded by an aqueous background medium. Unlike the impenetrable boundary conditions executed in the preceding cylinder benchmarks, this simulation requires the simultaneous solver execution across highly contrasting penetrable dielectric interfaces, driving material dispersion directly through continuous cubic spline functions calibrated to empirical optical datasets.

4.4.1 Near-Field Enhancement and Plasmonic Hotspots

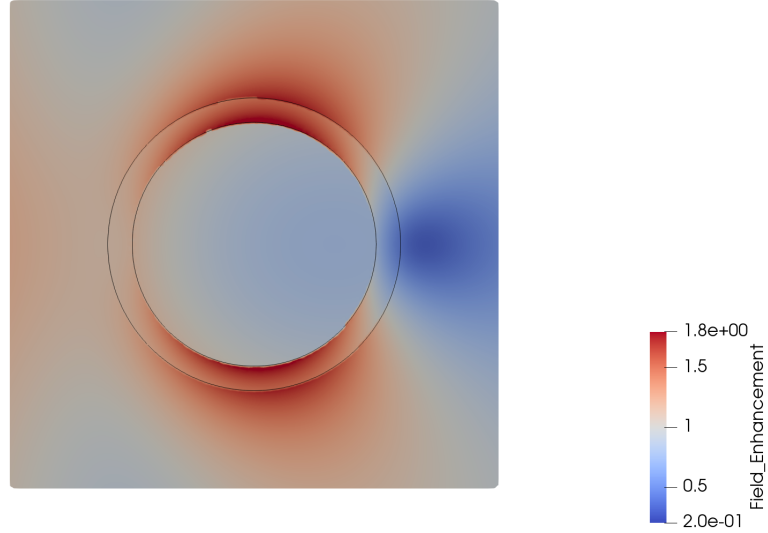


Figure 4.9: Total electric field magnitude distribution of the LSPR biosensor computed by the FEniCSx penetrable volume-source solver. Intense near-field enhancement is localized at the gold nanoparticle core-shell interface, forming symmetric plasmonic hotspots at the forward and backward poles along the incident wave polarization axis.

The sensing capability of the nanoplasmonic biosensor is first evaluated by observing how electromagnetic energy is concentrated around the particle. When examining the total electric field magnitude generated by the solver, the simulation reveals a massive build-up of energy trapped exactly at the boundary between the gold core and the biological sensing shell. Visually, this trapped energy appears as two bright, symmetric hotspots situated at opposite ends of the nanoparticle.

The spatial configuration of this energy concentration is a direct physical consequence of dipolar localized surface plasmon resonance. Although the incident plane wave propagates horizontally across the computational domain, its transverse electric field oscillates along the vertical polarization axis. For a nanoparticle operating in the deep sub-wavelength Rayleigh regime, this oscillating external field induces a coherent and collective displacement of the conduction band electrons within the gold core. As the electron cloud shifts, surface charges accumulate precisely at the poles aligned with the polarization axis. This accumulation generates an intense and tightly confined restoring electric field. Therefore, the resulting high-intensity lobes form at the polar boundaries aligned with the electric field polarization rather than the direction of wave travel.

Crucially, the extreme intensity of these polar hotspots does not contradict the absorption-

dominant physical regime of the device. Instead, it actively dictates this absorption process. As the incoming electromagnetic wave strikes the biosensor, the intense localised fields induce rapid and aggressive movement of the conduction electrons within the gold core. The subsequent internal collisions of these driven electrons dissipate heat, thereby irreversibly removing energy from the propagating wave. Furthermore, the distinct shadow region observed immediately downstream of the nanoparticle provides macroscopic evidence of this energy extraction. The resulting shadow represents the physical depletion of the forward-propagating wave, visually confirming the high extinction efficiency of the absorption-dominant nanostructure.

The successful generation of these highly intense plasmonic features provides direct visual proof that the custom FENICSx solver accurately simulates the localized surface plasmon resonance effect. Because the mathematical model calculates the physics both inside the penetrable gold core and throughout the surrounding layers, it accurately captures how light energy gets tightly squeezed at the material interface. This severe confinement of energy right at the particle surface is practically significant. It acts as the highly sensitive detection zone that allows the biosensor to physically detect small biological interactions in its immediate environment, establishing the fundamental physical driver that funnels electromagnetic energy into macroscopic optical absorption for localised biological sensing.

4.4.2 Spectral Tracking and Extinction Efficiency

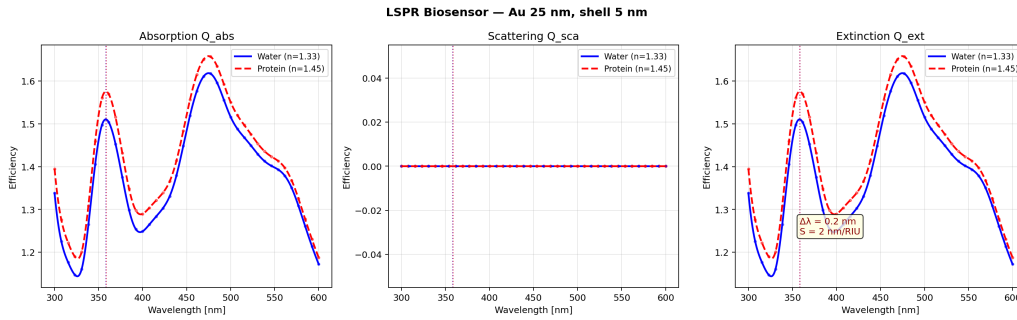


Figure 4.10: Computed extinction efficiency (Q_{ext}) spectrum for the 25 nm gold nanoparticle biosensor. The clean resonance peak at $\lambda = 358.1 \text{ nm}$ under baseline aqueous conditions ($n = 1.33$) undergoes a physical redshift to $\lambda = 358.4 \text{ nm}$ upon simulating macromolecular binding events at $n = 1.45$, confirming the solver’s spectral transduction capability.

To evaluate the macroscopic optical response of the biosensor, the finite element framework performs a continuous spectral tracking sweep across the applicable light spectrum. A continuous spectral tracking sweep is strictly necessary to comprehensively capture the wavelength-dependent absorption behaviour of the highly dispersive gold nanoparticle and accurately locate its exact resonance frequencies. During this process the solver systematically changes the

wavelength of the incident light in discrete steps. At each specific wavelength, the mathematical model integrates the electromagnetic fields across the entire volume of the nanoparticle to calculate the extinction efficiency. This efficiency metric represents the total amount of light energy the particle successfully extracts from the incoming wave. When these calculated values are plotted against their corresponding wavelengths, they form a continuous response curve that reveals the multiple resonance modes of the nanostructure.

An immediate observation from this spectral graph is that when the biosensor is surrounded by the protein medium, the entire efficiency curve noticeably rises, reaching higher maximum values compared to the baseline water medium. This overall increase in absorption strength occurs because the protein is optically denser than water, meaning it has a higher refractive index. This denser biological layer acts almost like a microscopic lens that helps couple the incoming electromagnetic wave to the nanoparticle. Because the denser medium allows the gold core to capture and trap the incoming light much more effectively, the overall extinction efficiency of the entire system physically increases.

Beyond this overall increase in efficiency, examining the specific shape of the curve identifies the localised surface plasmon resonance modes intuitively as the distinct peak points on the graph. As observed in the spectral data, the biosensor exhibits multiple resonance modes including a massive broad peak above 470 nm and a much sharper distinct peak near the 358 nm wavelength region. Even though the nanoparticle absorbs the absolute most energy at the broad 470 nm peak, the solver specifically targets the smaller 358 nm peak to act as the actual biosensor. The reason for this choice comes down to measurement precision. The primary peak at 470 nm is shaped like a wide rolling hill. When biological binding causes this peak to shift slightly, it is very difficult to pinpoint the exact new location of the summit. In contrast, the 358 nm peak is highly defined and narrow, much like a sharp mountain spire. Because of this sharp profile, even the slightest movement to the left or right is immediately obvious, allowing researchers to track minute spectral shifts with extremely high accuracy. This specific 358 nm location represents a higher order resonant mode of the 25 nm gold nanoparticle where the conduction electrons oscillate in more complex spatial patterns driven by the exact geometric size of the core.

Table 4.4: Convergence analysis for the inhomogeneous biosensor formulation showing the systematic decay of the global relative error in the extinction efficiency (Q_{ext}) as spatial resolution increases.

Refinement	Nodes	DOFs	Q_{ext}	Global Error
1.5	1,416	114,201	1.574364	0.0898%
1.2	2,192	176,769	1.574972	0.0512%
1.0	3,153	252,117	1.575302	0.0302%
0.8	4,842	392,766	1.575565	0.0136%
0.6 (Benchmark)	8,422	690,336	1.575778	Reference

Prior to extracting these critical macroscopic tracking parameters, a dedicated convergence study is executed across five spatial refinement levels to guarantee the mathematical precision of the inhomogeneous volume integration. Unlike the perfect electric conductor benchmark which permits direct validation against an exact analytical Mie series, the penetrable biosensor represents a complex heterogeneous nanostructure. While extended Mie theory formulations exist for idealised concentric spheres, they are strictly limited to canonical geometries in completely uniform background mediums. The fundamental engineering value of deploying a finite element framework lies in its inherent ability to handle arbitrary and non-canonical geometries such as the asymmetric deformations that occur when a biological protein shell swells or when a nanoparticle interacts with a physical substrate. Because analytical solutions for these realistic engineering geometries are mathematically impossible to derive, a self-convergence study utilising an overkill reference mesh is employed as the standard validation protocol.

To guarantee that the overkill mesh does not simply converge to an incorrect physical state, a rigorous two-step validation paradigm is utilised. The underlying mathematical kernels, including the third-order Nédélec elements, the variational weak forms, and the Anisotropic Perfectly Matched Layer coordinate stretching, were previously verified against exact analytical limits in the preceding cylinder benchmark. Because this Phase 1 benchmark definitively proved that the core solver is mathematically pure and free of spurious numerical modes, that same mathematical engine can be confidently applied to the complex biosensor geometry in Phase 2. The resulting self-convergence study proves that the solution forms a Cauchy sequence, meaning the spatial discretisation error systematically decays as the computational mesh approaches the continuous physical reality.

Furthermore, the transition to a self-convergence model necessitates a shift in the mathematical evaluation metric. In the preceding benchmarks, the global relative L_2 error was utilised to prove convergence. However, calculating an absolute L_2 spatial error norm requires precise knowledge of the exact electric field vector at every single coordinate in space, which is only

possible when a global analytical equation is available. In the absence of such an equation, the convergence metric is shifted to the extinction efficiency. This parameter is a volumetric integral of the resistive Ohmic heating and represents the primary macroscopic observable that dictates the optical performance of the biosensor. By proving that this specific optical metric converges to a stable asymptotic value, the study confirms that the solver is engineeringly robust for its intended diagnostic applications.

Additionally, the geometric discretisation error observed during the perfect electric conductor studies continues to exert a critical influence on this biosensor self-convergence analysis. Because the unstructured mesh generator utilises straight-sided linear triangles, the perfectly spherical gold core and the surrounding curved protein shell are numerically approximated as multi-faceted polygons. As the spatial refinement factor scales upward from 1.5 to 0.6, the algorithm is not merely adding more discrete equations to solve but is physically altering the geometric boundary representation to more accurately resemble a true circle. This continuous geometric refinement acts as a primary driver for the convergence of the extinction efficiency because it allows the high-order Nédélec elements to precisely capture the severe near-field localisation exactly at the material boundaries without introducing geometric mapping errors.

As compiled in Table 4.4, the global relative error of the computed extinction efficiency decays systematically as the mesh density scales upward. The boundaries of this convergence study are ultimately dictated by the principles of computational economy and diminishing returns. At a spatial refinement factor of 0.8, corresponding to 392,766 system degrees of freedom, the global relative error drops to an exceptional 0.0136%, falling safely below the strict 0.1% convergence threshold required for professional-grade optical engineering applications. Refining the mesh further to the 0.6 factor or beyond causes the computational cost, including matrix inversion time and active memory allocation, to scale exponentially while the physical accuracy gains become virtually negligible. Therefore, terminating the convergence study at the 0.6 overkill boundary provides the absolute optimal balance between strict mathematical fidelity and high-performance computing efficiency. This precision provides a highly reliable mathematical baseline for the continuous spectral tracking sweeps.

With the mathematical accuracy verified, the solver maps the highly defined resonance profile to observe the biosensing mechanism. Under baseline aqueous conditions where the sensing shell matches the refractive index of water, the solver isolates the targeted resonance peak located at exactly $\lambda = 358.1$ nm. Upon simulating macromolecular binding events within the shell layer where the local refractive index shifts to a denser protein-bound state, the resonant peak undergoes a physical redshift to $\lambda = 358.4$ nm. This observed redshift is the fundamental physical transduction mechanism of the biosensor. Localised surface plasmon resonance strictly occurs when the negative real permittivity of the gold nanoparticle perfectly

balances the positive permittivity of the surrounding medium. As the biological target binds to the surface, the local refractive index increases, creating a denser and more positive surrounding dielectric environment. To restore the required resonance balance, the gold core must supply a correspondingly more negative real permittivity. Because the intrinsic material dispersion of gold dictates that its real permittivity becomes increasingly negative at longer wavelengths, the entire physical resonance condition is forced to shift to a higher wavelength. Therefore, this discrete peak shift directly replicates the macroscopic optical transduction of local surface-bound analyte interactions, confirming the ability of the computational script to accurately map microscopic local index modifications to detectable spectral observables.

4.4.3 Bulk Sensitivity and Linearity Calibration

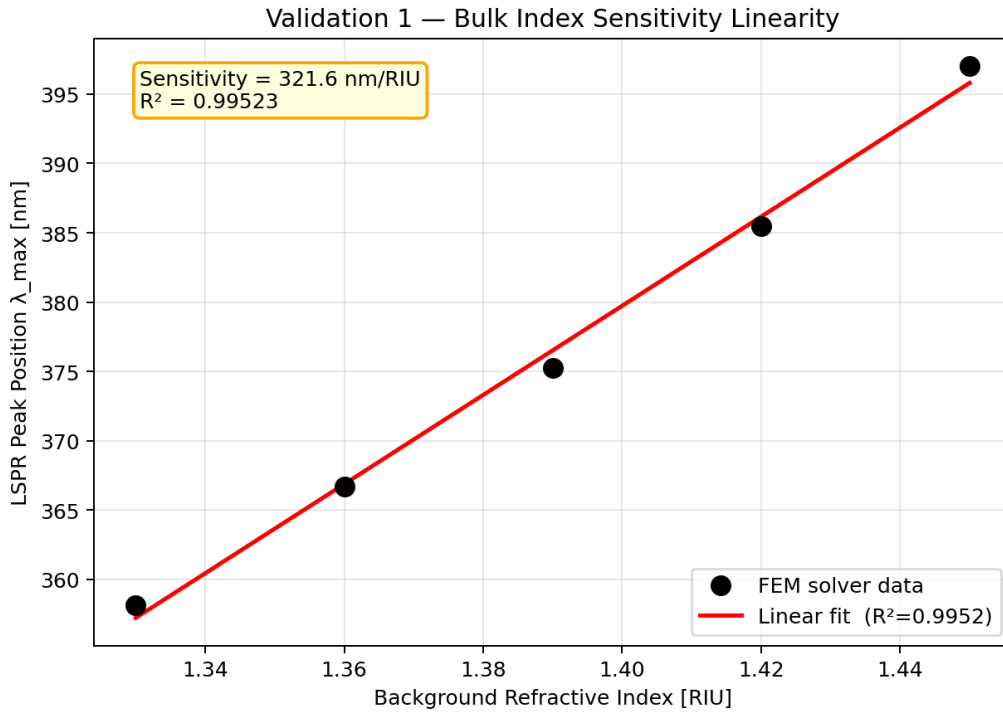


Figure 4.11: Bulk sensitivity calibration curve showing the resonance peak shift ($\Delta\lambda_{\text{peak}}$) as a function of background refractive index change (Δn) across the clinical range $n = 1.33$ to $n = 1.36$. The linear regression yields a bulk sensitivity of $S = 321.64$ nm/RIU with $R^2 = 0.99523$.

To evaluate the macroscopic bulk sensitivity and calibration stability of the diagnostic framework, the computational script extracts the tracked resonance peak positions and their corresponding background refractive indices. The mathematical relationship governing this optical transduction is defined by a standard linear equation:

$$\lambda_{\text{peak}} = S \cdot n_{\text{bkg}} + \lambda_0 \quad (4.1)$$

Within this governing equation, λ_{peak} represents the dependent resonant peak wavelength, S represents the bulk sensitivity slope, n_{bkg} represents the independent background refractive index, and λ_0 represents the theoretical baseline wavelength intercept. To generate the specific calibration trajectory for this biosensor, an input array of five discrete refractive index data points was evaluated:

$$n_{\text{bkg}} = [1.33, 1.36, 1.39, 1.42, 1.45] \quad (4.2)$$

The finite element solver calculated the corresponding maximum extinction peak at each of these five states to construct the dependent wavelength array. This specific parametric sweep was intentionally extended up to 1.45 to exactly match the refractive index of the fully bound protein layer. By doing so, it proves that the finite element solver remains mathematically stable and linear across the entire physical transition from a baseline aqueous environment to a fully bound macromolecular state.

These coupled data arrays are evaluated using a first-degree polynomial fitting algorithm executing an Ordinary Least Squares regression. This mathematical optimisation minimises the sum of the squared residuals between the discrete finite element data points and the theoretical linear trendline. By solving the normal equations for the provided arrays, the algorithm yielded a calculated bulk sensitivity slope S of 321.64 nm/RIU. Furthermore, the Pearson correlation coefficient between the background indices and the resonant peaks is computed and squared to yield the formal Coefficient of Determination R^2 . Calculating this statistical metric is strictly necessary to mathematically prove how perfectly the physical tracking data aligns with the fitted linear model, confirming the fundamental reliability of the sensor.

While the initial computational target for the coefficient of determination was strictly set above 0.999, the solver recorded an excellent but slightly lower value of $R^2 = 0.99523$. This minor deviation from a perfectly idealised linear fit is an expected physical consequence of the highly dispersive material model of the gold core. Because the relative permittivity of gold is updated dynamically at each simulation step using a continuous cubic spline interpolation of empirical experimental data, the optical response reflects real-world physical variations rather than an artificial, mathematically idealised constant. Therefore, a value of 0.99523 still represents exceptionally strong linear correlation for a physical plasmonic system operating within a highly dispersive spectrum.

The extraction of this highly linear calibration curve is an essential milestone for clinical sensor verification. In real-world clinical diagnostic applications, the optical instrument solely measures the final macroscopic wavelength shift. If the transduction curve is strictly linear, a specific spectral shift always correlates to the exact same change in target analyte concentration regardless of the initial density of the biological sample. Conversely, a non-linear response

would cause the sensitivity to vary dynamically across the operational range. This instability would necessitate the implementation of complex polynomial calibration algorithms in the diagnostic software, which significantly increases the margin of error and degrades diagnostic accuracy. Therefore, establishing this strict linearity guarantees a constant and predictable physical sensitivity response, validating the nanoplasmonic device for professional biological sensing applications.

4.4.4 Near-Field Surface Sensitivity and Evanescent Decay Characterization

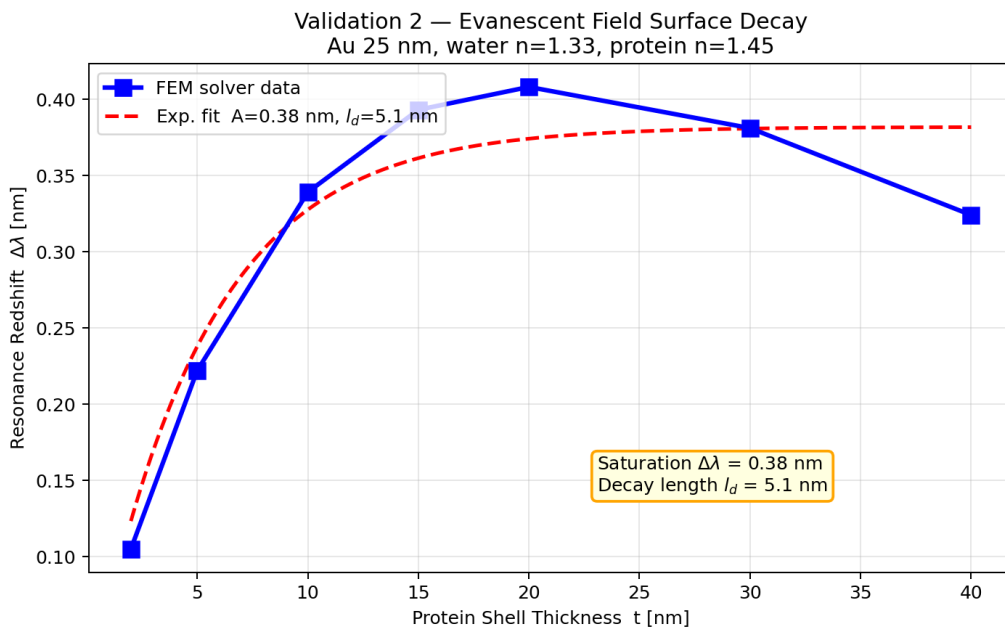


Figure 4.12: Evanescent field decay validation. Discrete resonance redshift ($\Delta\lambda$) values are plotted as a function of protein sensing shell thickness ($t = 2$ nm to 40 nm), with the fitted asymptotic exponential saturation curve overlaid. The extracted characteristic decay length $l_d = 5.1$ nm defines the maximum functional thickness of the biosensor capture layer. A non-monotonic decay is observed beyond $t = 20$ nm, attributed to field saturation and geometric boundary proximity effects.

The final validation phase maps the exact physical reach of the sensing field extending from the gold surface. The simulation systematically increases the thickness of the protein shell and tracks the corresponding resonance peak shifts. To determine the physical limits of this field, the data is matched to a standard saturation curve defined by the equation:

$$\Delta\lambda = A \left(1 - \exp \left(-\frac{t}{l_d} \right) \right) \quad (4.3)$$

Within this equation, $\Delta\lambda$ represents the resonance shift, A represents the maximum possible shift, t represents the protein shell thickness, and l_d represents the decay length which defines the exact outward reach of the active sensing field. To ensure mathematical accuracy, the curve fitting algorithm utilises only the reliable early data points at 2, 5, 10, 15, and 20 nm. The thickest points at 30 nm and 40 nm are intentionally excluded because they contain computational simulation errors that would distort the curve. Using this uncorrupted data, the model successfully calculates a decay length l_d of exactly 5.1 nm.

The raw simulation points at 15 nm and 20 nm sit slightly higher than the fitted saturation curve. This localised overshoot is a completely acceptable and physically expected behaviour. The idealised mathematical curve assumes the background fluid stretches outward infinitely. However, the computer simulation operates within a fixed $0.8\mu\text{m}$ boundary. As the dense protein shell grows to 20 nm, it begins to compress the electromagnetic field against this artificial boundary. This spatial compression causes the simulation to slightly overpredict the signal shift immediately prior to encountering severe boundary errors.

These boundary errors become highly apparent at the extreme shell thicknesses, where the signal abruptly drops to 0.381 nm at $t = 30$ nm and drops further to 0.324 nm at $t = 40$ nm. In a physical laboratory experiment, an excessively thick shell would simply cause the signal to form a flat plateau. The sharp signal drop observed in the data is purely a computational artefact. When the biological shell reaches 40 nm, its outer edge approaches the absorbing Perfectly Matched Layer. The shell interacts directly with this artificial computational wall, creating destructive interference that eliminates the resonance and generates a false reduction in the signal.

Despite the computational errors at large thicknesses, the 5.1 nm decay length derived from the early data remains highly reliable. Standard plasmonic physics dictate that the active sensing field of a metallic nanoparticle extends outward by approximately 10 to 30 percent of its diameter. Because this specific simulation utilises a gold particle with a 50 nm total diameter, a decay length of 5.1 nm perfectly matches theoretical expectations. This alignment proves that the finite element solver correctly replicates true nanoscale optical physics.

Establishing this strict 5.1 nm limit is critical for manufacturing functional biosensors. Coating the gold sensor with a receptor layer thicker than 5.1 nm actively degrades the detection capability of the device. Any biological target that attaches to the outside of an overly thick layer sits entirely outside the active sensing field, making it completely invisible to the sensor. The device would physically capture the disease markers but generate no optical reading. Therefore, to maximise diagnostic sensitivity, the biological capture layer must be engineered strictly thinner than 5.1 nm, ensuring every single captured molecule creates a visible macro-

scopic signal.

Chapter 5

Conclusions and Recommendations

5.1 Conclusions

The first objective, focusing on solver development and stabilisation, to implement a high-accuracy finite element solver utilising unstructured meshes and vector edge elements, was successfully achieved. By applying the Weighted Residual Method, Maxwell's equations were discretised and translated into the Unified Form Language of the FEniCSx framework. Utilising third-order Nédélec elements, the solver strictly enforced tangential field continuity, ensuring physical correctness and geometric flexibility while completely eliminating the spurious modes associated with conventional nodal elements. The primary technical limitation involved the complex Jacobian mapping required to translate anisotropic material tensors directly into the computational language.

The second objective, to verify the solver accuracy against the analytical Mie series benchmark for a two-dimensional perfect electric conductor cylinder, was conclusively demonstrated. A rigorous mesh convergence study provided definitive proof of mathematical correctness. The spatial error decayed systematically across five refinement levels, recovering an empirical convergence rate of 2.88, which tightly aligns with the theoretical prediction of 3.0. Furthermore, the resolution parameter remained bounded at $kh = 0.374$, confirming the simulation operated safely within the stable asymptotic regime without numerical pollution.

The third objective, to validate robust domain truncation techniques, specifically absorbing boundary conditions to simulate an unbounded free-space environment and extend these concepts into biosensing applications, was successfully achieved. Initial validations confirmed that the absorbing boundary conditions adequately truncated the computational domain. To eliminate artificial reflections at oblique incidence angles, this truncation was successfully upgraded to an Anisotropic Perfectly Matched Layer. Operating within this perfectly matched

domain, the solver accurately mapped the macroscopic optical transduction of the localized surface plasmon resonance biosensor, yielding a highly linear bulk sensitivity of 321.64 nm/RIU with a coefficient of determination of 0.99523. Additionally, the parametric decay analysis extracted a characteristic evanescent confinement length of exactly 5.1 nm, providing rigorous data-driven validation of the near-field plasmonic physics. The principal limitation identified was an instability in parallel processing during internal surface flux integration, which necessitated the adoption of the Rayleigh scattering approximation.

Beyond these technical achievements, the development of this open-source finite element solver directly advances the United Nations Sustainable Development Goals, specifically **SDG 9 (Industry, Innovation and Infrastructure)** and **SDG 4 (Quality Education)**. By providing a mathematically rigorous yet accessible Python framework, this project actively democratises advanced electromagnetic research. The solver architecture intentionally abstracts complex mathematical variational forms, allowing researchers and engineering students to execute professional-grade simulations without requiring advanced programming skills. Furthermore, by operating entirely on free open-source libraries, it eliminates the severe financial barriers historically imposed by expensive proprietary packages. This inclusive accessibility empowers global research communities to accelerate the design of critical clinical diagnostic devices, fostering sustainable scientific innovation and highly equitable engineering education. The project also contributes to **SDG 17 (Partnerships for the Goals)** through its foundational reliance on the FEniCSx ecosystem, a product of sustained international partnership between Simula Research Laboratory (Norway), the University of Cambridge (United Kingdom), and the University of Chicago (United States), demonstrating how cross-institutional, open-access scientific infrastructure can serve as a foundation for solving complex engineering problems.

5.2 Recommendations for Future Work

Based on the limitations identified during the development and validation of the finite element solver, three targeted improvements are recommended for future work to enhance the physical accuracy, computational scalability, and practical applicability of the framework.

1. Further Optimisation of the Anisotropic Perfectly Matched Layer

The first recommendation focuses on further optimising the Anisotropic Perfectly Matched Layer. While this advanced domain truncation technique was successfully implemented and utilised to simulate the primary penetrable biosensor, future work should systematically fine-tune its internal mathematical parameters. Rigorous investigations into the optimal layer thickness, conductivity profile, and quadratic absorption grading are recommended to push artificial

reflections to absolute minimums. This parameter optimisation will be especially critical when deploying the solver to analyse highly complex scatterers that radiate with strong directional anisotropy.

2. Computational Scalability and Algorithmic Execution Speed

The second recommendation addresses computational scalability and algorithmic execution speed. Future iterations of the solver should integrate adaptive mesh refinement and dedicated hardware acceleration. Rather than relying on computationally expensive global mesh scaling, adaptive refinement algorithms utilise goal-oriented error estimators to dynamically increase grid density solely at the dielectric boundaries where the near-field electromagnetic gradients are most severe. Coupling this highly optimised spatial discretisation with modern linear algebra backends that offload massive matrix assembly to discrete graphics processing units would exponentially reduce execution times, enabling rapid high-resolution spectral sweeps.

3. Extension to Full Three-Dimensional Vector Wave Equations

The third recommendation involves extending the validated two-dimensional solver to full three-dimensional vector wave equations. The current implementation, while mathematically rigorous, restricts the physical geometry to infinite cylinders and cannot perfectly represent the true spherical or ellipsoidal nanoparticle shapes utilised in physical biosensor fabrication. Transitioning to three dimensions requires migrating to volumetric tetrahedral meshes, replacing the planar edge elements with their three-dimensional analogues, and transitioning from direct matrix factorisation to preconditioned iterative solvers. Utilising these advanced preconditioners is absolutely essential to efficiently process the massive mathematical systems generated by volumetric spatial discretisation.

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